

Cohesive Closure Repair and Its Hodge, Kernel, and Projection Shadows:

From Nonlinear Defects to Tangent Semigroups, with the Physical-Action Boundary

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Abstract

Modal Triplet Theory repeatedly uses fixed points, positive Hessians, heat kernels, coherent projectors, and perturbative graphs. These objects have often appeared as neighboring constructions without one theorem stating their exact common source. This paper supplies that theorem at the structural tier. Begin with a nonlinear defect section Φ on a metric configuration space and the positive repair cost $\mathcal{C} = \frac{1}{2} \|\Phi\|^2$. Its negative gradient is closure repair. At a zero-defect fixed point the Hessian is exactly the normal operator $D\Phi^\dagger D\Phi$; its heat semigroup, Green operator, spectral projectors, local kernel, and complete finite repair-vertex jet are then different mathematical shadows of the same linearized source. At a nonzero-defect critical point an additional covariant residual term is present, so the normal-square formula cannot be used without qualification.

The construction is extended to local Čech descent and to an α -twisted cohesive module. The projective twist cancels in the endomorphism algebra, the Maurer–Cartan curvature $F(a) = da + a^2$ gives a canonical nonlinear repair residual, and a chosen Hermitian structure yields the Hodge operator and tangent heat flow. An exact stratified example shows simultaneously that the full cohesive operator may remain regular while every uniform fiberwise Green operator diverges across a cohomology-rank jump. Compatible isometric intertwiners transport the repair flow, functional calculus, projectors, kernels, and formal graph coefficients.

The paper also proves the necessary boundary. A positive repair square is not a signed physical action. It loses first-order sign, phase, Morse index, causal support, and generally the physical interaction vertices. A direct action requires a variational anchor satisfying Helmholtz conditions; a universal multiplier lift always exists; and a cyclic Maurer–Cartan source admits a Chern–Simons-type signed action. For the q79 program, the cohesive Hartshorne–Serre benchmark closes the structural chain exactly after a Hermitian choice. Selection of the physical V_3/W_9 endpoint, its HYM metric, full Hull–Strominger action, Lorentzian domains, and the continuum-to-finite intertwiner remain open. Thus closure repair explains why the operator and kernel language should recur, without promoting that recurrence into a completed theory of dynamics or quantization.

Version 8 Revision Note

Supersedes The local Version 7 manuscript; no public release is implied.

Reason Local nested transfer needed its global overlap and actual q79 rational-topology context.

Resolution Imports the curved projective-overlap theorem and the 96-to-88 Hirsch contraction, with examples explaining cyclic-pairing compensation, strict topology and nontrivial coefficient interactions.

Retained result Earlier charge, residual, projection and transfer results keep their stated scopes. Fixed Points I–VI are unchanged.

Remaining boundary Topological contraction and algebraic descent do not select a physical metric, endpoint, analytic reduction or 27-state map.

Version 7 Revision Note

Supersedes The local Version 6 manuscript; no public release is implied.

Reason The inner-charge discussion and finite transfer results needed their nested Witten and noncentral coefficient constructions explained together.

Resolution Integrates six local records from outer nilpotence through restricted contraction to a positive curved coefficient algebra. Distinguishes the self-adjoint coefficient matrix from the full module superconnection, cyclic from positive pairings, and minimal from nonminimal transfer.

Retained result Previous residual, projection, first-jet, charge and finite-response results remain at their stated scopes. Fixed Points I–VI are unchanged.

Remaining boundary The finite tensor model does not select a physical factorization, global shared-line realization, analytic domain, action or value.

Version 6 Revision Note

Supersedes The local Version 5 manuscript; no public release is implied.

Reason The common-source discussion had not yet integrated the single-charge, curved-charge, central-curvature and admissible-tangent results.

Resolution Explains six finite/local source records: shared central characters, nilpotent factorization, phase-dependent curved squares, complementary Jordan/Lie readouts, central identifiability and intrinsic stability on a constrained phase circle.

Retained result The previous residual, action, projection, finite transfer and first-jet qualifications remain. Fixed Points I–VI are unchanged.

Remaining boundary An algebraic charge is not an electric charge; phase dependence is not unique phase recovery; a small center does not select physical constants. The physical source, admissible stratum, domains and shared-line realization still require their own construction.

Version 5 Revision Note

Supersedes The local Version 4 manuscript; no public release is implied.

Reason Compatible projection alone did not distinguish compression, inverse-weighted elimination and higher-product transfer, or explain the additional harmonic modes required by finite covariance completion.

Resolution Integrates eleven source records into one comparison: leakage, finite gauge descent, the 36- and 144-dimensional calculi, and the 48-dimensional response retract with its nontruncating higher products. Separates a harmonic quotient from a flat degeneration and an all-arity theorem from exhaustive operation tables.

Retained result The residual, action and projector-completeness corrections remain. No foundational Fixed Points theorem is changed.

Remaining boundary Finite covariance and higher operations do not select a physical HYM endpoint, continuum intertwiner, action or measured coupling.

Version 4 Revision Note

Supersedes The local Version 3 manuscript; no public release is implied.

Reason An older continuum-to-finite contract bounded the whole ambient harmonic projector using only an intertwining residual on the retained space.

Resolution Gives an exact counterexample, separates retained from ambient spectral sectors, and states the completeness and isolated-cluster conditions needed for projector comparison.

Retained result Exact reducing intertwiners still transport every retained spectral projector. The repair, residual and finite numerical results are unchanged.

Remaining boundary No additional ambient zero modes, common analytic domains and perturbative spectral separation require their own certificates.

Version 3 Revision Note

Supersedes The local Version 2 manuscript; no new public release is implied.

Reason The physical residual had only a schematic description, and the projection statement did not distinguish a unitary equivalence from an isometric embedding with transverse coupling.

Resolution Displays the six additional residuals, their derivatives and the corrected 25-block rank-102 support. States the extra adjoint condition for embedded Hessian transport, explains transported metrics, and organizes the physical endpoint contract into three structured source types.

Retained result The residual-Hessian, cohesive, descent, repair-jet and action-boundary theorems remain at their stated tiers. Previously established local and finite results are not reopened.

Remaining boundary A structural support mask is not a numerical matrix; symmetry reduction requires endpoint equivariance. Physical source selection, the common analytic domain and the four-dimensional action remain distinct.

Version 2 Revision Note

Supersedes Version 1. The previous release remains public until a new release is approved.

Reason Connect strict-gradient repair to the finite-mode recurrence obstruction.

Resolution Connects strict repair with the stationary-ensemble and finite-mode recurrence obstructions.

Retained result Correctly scoped results and explanatory examples remain; no physical source-selection claim is promoted.

Remaining boundary Realization hypotheses and physical source or apparatus bridges remain separate obligations. This is an unreleased authoring revision.

Version 1 revision note

Supersedes. No earlier paper. Version 1 establishes the baseline for this new theorem-owner paper.

Reason. The Fixed Points, wave-particle, and diagrammatics papers each use a downstream part of the same mathematical chain, but none is the proper owner of the upstream derivation from nonlinear closure repair.

Resolution. Version 1 proves and explains the structural chain

nonlinear closure defect $\longrightarrow$ repair flow $\longrightarrow$ linearized Hodge operator,
linearized Hodge operator $\longrightarrow$ kernel and projection shadows.

It consolidates separately certified theorem packets, proves their composition in standalone form, includes the nonzero-defect correction, and separates the positive repair functional from a signed physical action.

Retained result. The exact component theorems and their executable witnesses are retained at their established tiers. Their ownership is reorganized rather than weakened: this paper owns the composition, while the downstream papers retain their fixed-point, wave-particle, and graph theorems.

Remaining boundary. No observed mass, coupling, mixing angle, cosmological datum, or fitted coefficient enters the theorems. The paper does not prove a Lorentzian action, interacting QFT, ultraviolet-complete quantum gravity, unique actualization, or selection of the physical q79 endpoint. The q79 cohesive application remains an exact structural benchmark rather than a physical promotion.

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1 The question before the operator

Many physical formalisms begin with an operator. One specifies a Hamiltonian, a Laplacian, a Dirac operator, a Hessian, or a propagator and then asks what it does. MTT begins one question earlier:

What is being repaired, and why should its first observable shadow be an operator at all?

The answer developed here is deliberately modest. Suppose a state is admissible when a nonlinear defect vanishes. The squared size of the defect is then a natural nonnegative cost. Its gradient points in the direction in which inadmissibility grows most rapidly; its negative gradient therefore defines a repair flow. Linearizing that flow at an exact admissible state produces a positive normal operator. Once that operator exists, the heat semigroup, Green operator, spectral projectors, and local kernels belong to ordinary functional analysis.

This does not make every physical operator a theorem of MTT. It identifies one precise mechanism by which a large and important class of operators can arise. The mechanism has hypotheses, and several of them carry physical content: the configuration space, defect bundle, metric, connection, domain, background, and interpretation must all be selected. Hiding any one of these choices inside the word “closure” would merely relocate the input.

The central distinction of this paper is therefore:

The repair operator is derived from a selected defect and metric. The selected defect and metric are not derived by notation.

1.1 How to read the paper

Readers interested in the universal differential geometry may read [sections 2](#) and [3](#). The cohesive and q79 construction is in [sections 4](#) and [14](#). The operator, kernel, and projection chain is proved in [sections 6](#) and [7](#). The relation to perturbative graphs is in [section 11](#). The crucial warning that repair is not automatically action appears in [section 12](#). The final status and theorem-ownership tables are in [sections 16](#) and [18](#).

1.2 Vocabulary of claim strength

We use four labels throughout.

1. **Exact** means proved from the displayed assumptions.
2. **Exact structural benchmark** means proved on the named q79 object, but not promoted to the selected physical endpoint.
3. **Conditional transport** means the conclusion follows if the stated chain, metric, or transform intertwiner exists.
4. **Open physical promotion** means that the required source object or comparison has not been constructed and certified.

These labels are part of the mathematics. They prevent a theorem about an available representation from being mistaken for a theorem selecting the representation realized in nature.

2 Residual geometry and the repair Hessian

2.1 The nonlinear source

Let $(\mathcal{M}, g)$ be a finite-dimensional Riemannian manifold. The same formulas extend to a Hilbert manifold when the derivatives, adjoints, and domains used below exist. Let

$$(\mathcal{E}, h, \nabla^\mathcal{E}) \longrightarrow \mathcal{M}$$

be a real metric or complex Hermitian vector bundle with metric connection, and let

$$\Phi \in \Gamma(\mathcal{E})$$

be a twice differentiable defect section. In the complex case every scalar pairing below means its real part. Define

$$\mathcal{C}(p) = \frac{1}{2}h(\Phi(p), \Phi(p)). \tag{2.1}$$

At $p \in \mathcal{M}$, write

$$r_p = \Phi(p), \tag{2.2}$$

$$J_p X = \nabla_X^\mathcal{E} \Phi, \tag{2.3}$$

$$B_p(X, Y) = \nabla_X^\mathcal{E} \nabla_Y^\mathcal{E} \Phi - \nabla_{\nabla_X^\mathcal{M} Y}^\mathcal{E} \Phi. \tag{2.4}$$

The tensor B_p is the covariant second derivative of the defect.

Theorem 2.1 (Covariant residual-Hessian identity). *For every $X, Y \in T_p \mathcal{M}$,*

$$d\mathcal{C}_p(X) = h(r_p, J_p X), \tag{2.5}$$

$$\text{Hess}_p \mathcal{C}(X, Y) = h(J_p X, J_p Y) + h(r_p, B_p(X, Y)). \quad (2.6)$$

Equivalently,

$$\text{grad } \mathcal{C}(p) = J_p^\dagger r_p, \quad H_p = J_p^\dagger J_p + S_{r_p}, \quad (2.7)$$

where

$$g(S_{r_p} X, Y) = h(r_p, B_p(X, Y)).$$

The residual form is symmetric, including when $\nabla^\mathcal{E}$ has curvature.

Proof. Metric compatibility gives

$$X[\mathcal{C}] = h(\nabla_X^\mathcal{E} \Phi, \Phi),$$

which proves (2.5). Differentiate once more and subtract the Levi-Civita correction:

$$\begin{aligned} \text{Hess } \mathcal{C}(X, Y) &= X[\text{d}\mathcal{C}(Y)] - \text{d}\mathcal{C}(\nabla_X^\mathcal{M} Y) \\ &= h(\nabla_X^\mathcal{E} \Phi, \nabla_Y^\mathcal{E} \Phi) + h\left(\Phi, \nabla_X^\mathcal{E} \nabla_Y^\mathcal{E} \Phi - \nabla_{\nabla_X^\mathcal{M} Y}^\mathcal{E} \Phi\right). \end{aligned}$$

This is (2.6). The antisymmetric part of B is $R^\mathcal{E}(X, Y)\Phi$. Curvature of a metric or Hermitian connection is skew-adjoint, so $\text{Re } h(\Phi, R^\mathcal{E}(X, Y)\Phi) = 0$. The residual term is therefore symmetric. $\square$

Corollary 2.2 (Exact-zero normal square). *If p_* is an exact closure state, $\Phi(p_*) = 0$, then*

$$\text{grad } \mathcal{C}(p_*) = 0, \quad \text{Hess}_{p_*} \mathcal{C} = J_*^\dagger J_* \geq 0. \quad (2.8)$$

Moreover,

$$\ker \text{Hess}_{p_*} \mathcal{C} = \ker J_*.$$

This is the exact source of the positive repair Hessian used later. It is also the point at which a common shortcut must stop. Equation (2.8) is not the general Hessian formula.

Corollary 2.3 (Nonzero-defect correction). *A state p is critical precisely when*

$$J_p^\dagger r_p = 0.$$

If $r_p \neq 0$, the Hessian is

$$J_p^\dagger J_p + S_{r_p},$$

and the second term generally cannot be reconstructed from r_p and J_p alone. On $\ker J_p$, local stability is controlled entirely by S_{r_p} .

Example 2.4 (Why the correction cannot be dropped). Let $\Phi_a : \mathbb{R} \rightarrow \mathbb{R}^2$ be

$$\Phi_a(x) = (x, x^2 + a), \quad a = 1.$$

At $x = 0$, the residual is $(0, 1)$, the derivative is $(1, 0)$, and $J^\dagger r = 0$. Thus $x = 0$ is critical although the defect is nonzero. The Gram term is $J^\dagger J = 1$, the residual correction is 2, and

$$\text{Hess}_0 \mathcal{C} = 3.$$

Using only the normal square would give the wrong local stiffness by a factor of three.

2.2 Symmetry and physical zero modes

Let a Lie group G act isometrically on $\mathcal{M}$ and fiberwise isometrically on $\mathcal{E}$, and suppose Φ is equivariant.

Proposition 2.5 (Orbit zero modes). *If p is critical and $\xi_{\mathcal{M}}(p)$ is tangent to its G -orbit, then*

$$\text{Hess}_p \mathcal{C}(\xi_{\mathcal{M}}(p), Y) = 0$$

for every $Y \in T_p \mathcal{M}$.

Proof. Invariance gives $d\mathcal{C}(\xi_{\mathcal{M}}) = 0$ identically. Differentiate in the direction Y . The term containing $d\mathcal{C}(\nabla_Y \xi_{\mathcal{M}})$ vanishes at a critical point, leaving the claimed Hessian identity. $\square$

Consequently, positivity in a gauge or phase-symmetric system means positivity on a slice normal to the symmetry orbit. Requiring strict positivity on the unreduced space would incorrectly reject the zero modes that encode the symmetry.

3 Repair as a flow

3.1 Dissipative and reversible parts

Pure repair is the negative-gradient equation

$$\frac{du}{ds} = -\text{grad } \mathcal{C}(u). \quad (3.1)$$

The parameter s orders stabilization. It is not automatically Lorentzian time. A more general reversible–dissipative split is

$$\frac{du}{ds} = (K - \Gamma) \text{grad } \mathcal{C}(u), \quad (3.2)$$

where K is skew-adjoint and Γ is self-adjoint and positive semidefinite.

Theorem 3.1 (Repair monotonicity and tangent generator). *Along every solution of (3.2),*

$$\frac{d}{ds} \mathcal{C}(u(s)) = -g(\text{grad } \mathcal{C}, \Gamma \text{grad } \mathcal{C}) \leq 0. \quad (3.3)$$

At a critical point p , the linearized generator is

$$D[(K - \Gamma) \text{grad } \mathcal{C}]_p = (K_p - \Gamma_p) H_p^\sharp. \quad (3.4)$$

At an exact-zero state and for pure repair, the tangent equation is

$$\partial_s v = -J_*^\dagger J_* v, \quad v(s) = e^{-s J_*^\dagger J_*} v(0). \quad (3.5)$$

Proof. Pair (3.2) with $\text{grad } \mathcal{C}$. The K -term vanishes by skew-adjointness, and positivity of Γ gives (3.3). At a critical point the derivatives of K and Γ multiply $\text{grad } \mathcal{C}(p) = 0$, so only the Hessian term remains. The exact-zero specialization follows from [theorem 2.2](#). $\square$

Corollary 3.2 (Strict-dissipation circulation no-go). *If Γ is positive definite along a periodic orbit of (3.2), that orbit is constant. A persistent circulation therefore requires a null-dissipation sector, a conservative sector, a relative equilibrium generated by symmetry, or external driving.*

This small result matters conceptually. A particle-like phase circulation cannot be obtained merely by drawing a loop around a strictly dissipative minimum. Repair explains stabilization. Reversible phase transport requires additional selected structure.

This also separates repair from stationary noise. Fixed Points III owns the stationary-ensemble obstruction for strict Lyapunov repair: under its integrability and invariance hypotheses, a stationary ensemble has zero dissipation almost everywhere. A closed finite-spatial-mode unitary model with compact resolvent supplies a different obstruction, namely recurrence rather than irreversible mixing. Removing only an occupation-number cutoff does not remove that finite-spatial-mode recurrence.

Thus a stationary diffusion limit needs an independently justified recurrent fast driver and a mixing/scaling argument, a continuum or thermodynamic limit, or a declared open system. Toral-chaos examples remain valid examples of separately supplied drivers; they are not flows of a strictly decreasing repair cost. These distinctions preserve the repair Hessian theorem and locate the extra dynamical input precisely. The owning discussion and certificates are in Fixed Points III and the [curated reproduction repository](#), under the finite-mode recurrence and fixed-point noise results; their proofs are not duplicated here.

3.2 Fixed points and basins

The flow may possess many fixed points or many attraction basins. Their existence, uniqueness, regularity, and convergence require the familiar compactness, coercivity, analytic-semigroup, or Łojasiewicz–Simon hypotheses developed in Fixed Points I [24]. The present paper does not duplicate those theorems. It supplies the upstream residual from which one eligible flow may be formed.

If a compact connected sector contains no zero of Φ , the continuous cost $\mathcal{C}$ attains a positive minimum. Such a minimum is a mathematically precise persistent-defect candidate. It is not automatically a particle: its physical interpretation still depends on sector selection, stability modulo symmetry, reversible transport, and the map to observables.

4 The cohesive source

4.1 Why a complex rather than only a bundle

The q79 source program contains an α -twisted Hartshorne–Serre object

$$\mathcal{S}_{\text{HS}} \in D^b(J, \alpha)$$

with a bounded local perfect atlas. Its open-stratum contraction uses an inverse transverse coordinate and cannot pass through the exceptional stratum. Discarding the full complex would therefore erase precisely the place at which its cohomology rank changes.

Cohesive modules provide a lawful alternative. A bounded α -twisted perfect atlas is represented by an α -twisted graded smooth module $E^\bullet$ with an integrable antiholomorphic superconnection

$$\overline{\mathbb{E}} = \overline{\mathbb{E}}_0 + \overline{\mathbb{E}}_1 + \overline{\mathbb{E}}_2 + \cdots, \quad \overline{\mathbb{E}}^2 = 0. \quad (4.1)$$

This is standard cohesive and twisted-complex mathematics [46, 3, 48, 49, 2, 4]. The q79-specific statement is that the certified local atlas satisfies its hypotheses.

Theorem 4.1 (Endomorphism untwisting). *Let the local projective transitions of $E^\bullet$ satisfy*

$$G_{ij}G_{jk}G_{ki} = \alpha_{ijk}\text{Id}.$$

Then $\text{End}(E^\bullet)$ glues as an ordinary global graded algebra by conjugation. The commutator differential

$$d_{\text{End}}(T) = [\overline{\mathbb{E}}, T] \quad (4.2)$$

satisfies $d_{\text{End}}^2 = 0$.

Proof. On a triple overlap,

$$\text{ad}(G_{ij}) \text{ad}(G_{jk}) \text{ad}(G_{ki}) = \text{ad}(\alpha_{ijk} \text{Id}) = \text{Id}.$$

Thus the scalar gerbe cocycle disappears from the endomorphism transition. The graded Jacobi identity and $\overline{\mathbb{E}}^2 = 0$ give

$$d_{\text{End}}^2(T) = [\overline{\mathbb{E}}^2, T] = 0.$$

□

The twist has not vanished from the object. It has become invisible to the adjoint algebra. This is exactly what is needed for a global deformation complex without falsely trivializing the original gerbe.

4.2 The canonical nonlinear residual

Set

$$\mathcal{A} = \Omega^{0,*}(\text{End } E), \quad d = [\overline{\mathbb{E}}, -].$$

For a degree-one perturbation $a \in \mathcal{A}^1$, define

$$F(a) = da + a^2 = (\overline{\mathbb{E}} + a)^2. \quad (4.3)$$

Then $F(a) = 0$ precisely when the perturbed superconnection remains integrable. This is not an arbitrary polynomial chosen to resemble an interaction. It is the curvature of the perturbed cohesive source.

Choose a Hermitian structure and a gauge row. On degree one, let

$$\Phi_{\text{MC}}(a) = (F(a), d_0^\dagger a), \quad \mathcal{C}_{\text{MC}}(a) = \frac{1}{2} \|F(a)\|^2 + \frac{1}{2} \|d_0^\dagger a\|^2. \quad (4.4)$$

Theorem 4.2 (Cohesive Maurer–Cartan repair). *At the integrable background $a = 0$,*

$$D\Phi_{\text{MC}}(0) = (d_1, d_0^\dagger), \quad (4.5)$$

and

$$\text{Hess}_0 \mathcal{C}_{\text{MC}} = d_1^\dagger d_1 + d_0 d_0^\dagger = \Delta_1. \quad (4.6)$$

The tangent negative-gradient flow is

$$\partial_s a = -\Delta_1 a, \quad a(s) = e^{-s\Delta_1} a(0).$$

The nonlinear curvature also satisfies the Bianchi identity

$$d_a F(a) = 0, \quad d_a = d + [a, -].$$

Proof. Differentiate (4.3) at zero. The quadratic term has zero first derivative, so $DF(0) = d_1$. Apply theorem 2.2 to the two residual rows in (4.4). The Bianchi identity follows by expanding $dF + [a, F]$, using $d^2 = 0$ and the graded Leibniz rule. □

The algebraic residual is canonical once the cohesive object is fixed. The Hermitian metric, adjoint, normalization, and physical moment-map rows are not. In particular, $d_0^\dagger a = 0$ is a gauge slice and must not be renamed the physical HYM equation.

5 One charge, complementary curvature readouts

The preceding construction starts from a defect and a metric. The next question is whether its reversible, dissipative and curvature operators can come from fewer independent choices. Six finite/local source records give a precise partial answer. Here *charge* means an algebraic operator, as in a differential or supersymmetry charge, not an electric charge or a selected particle quantum number. The Hodge-square mechanism has established precedents in supersymmetric quantum mechanics [51]. Its use here supplies a constrained source architecture, not a claim that the mechanism itself is new.

5.1 The shared center must be specified as a representation

On a qutrit basis e_j , $j \in \mathbb{Z}_3$, put

$$Xe_j = e_{j+1}, \quad Ze_j = \omega^j e_j, \quad \omega^3 = 1, \quad \omega \neq 1.$$

Then $XZ = \omega^{-1}ZX$ and $XZX^{-1}Z^{-1} = \omega^{-1}I$. The Fourier matrix with entries ω^{jk} conjugates X to Z and Z to X^{-1} . On $(\mathbb{C}^3)^{\otimes 3}$, placing X, Z separately in each factor gives three commuting lanes with the same scalar commutator $\omega^{-1}I_{27}$ [33].

The representation qualification matters. These are independent lane actions with their scalar centers identified, a central-product construction. They are not the diagonal tensor action of one qutrit group. For that diagonal action the two operators $X^{\otimes 3}, Z^{\otimes 3}$ commute, since their commutator is $\omega^{-3}I_{27} = I_{27}$. Thus even the same tensor dimension does not specify the central character. The witness does not identify this carrier with CLN, the physical finite 27-dimensional carrier, or an exceptional Jordan algebra. Nor does an order-three scalar action alone construct the shared differential line with its continuum connection and holonomy.

5.2 Nilpotence links reversible transport and damping

Let d act on a finite-dimensional complex Hilbert space, with $d^2 = 0$. For a unit phase ζ , define

$$B_\zeta = \zeta d + \bar{\zeta} d^\dagger, \quad \Delta = dd^\dagger + d^\dagger d.$$

The square expansion gives $B_\zeta^\dagger = B_\zeta$, $B_\zeta^2 = \Delta \geq 0$, and $\ker \Delta = \ker d \cap \ker d^\dagger$. Let P_h project onto this harmonic space. If Z_h is skew-adjoint and commutes with d and $d^\dagger$, then for $\gamma \geq 0$ the generator

$$\mathcal{L} = -iB_\zeta - \gamma\Delta + Z_h P_h$$

has mutually commuting summands. Consequently

$$e^{s\mathcal{L}} = e^{sZ_h P_h} e^{-isB_\zeta} e^{-\gamma s\Delta}, \quad \frac{d}{ds} \|\psi\|^2 = -2\gamma \langle \psi, \Delta \psi \rangle.$$

This is the single-charge factorization result [38]. It derives compatibility between the reversible operator and damping from d , while leaving d , its metric, ζ , γ , and Z_h as supplied data. The parameter s is not automatically physical time. Continuum extensions need domains and strong commutation, not just formal commutator identities.

For a small example take $d = E_{12}$ on $\mathbb{C}^3$. Then $\Delta = \text{diag}(1, 1, 0)$. The first two modes decay for $\gamma > 0$; on the third mode both B_ζ and Δ vanish. Persistent harmonic phase motion must therefore be supplied by $Z_h P_h$, not by the Hodge operator. This is compatible with the

earlier strict-gradient circulation obstruction: the undamped motion belongs to the extra skew generator, not to gradient repair.

Changing ζ changes the unsquared operator in a fixed frame, but that is not by itself an observable distinction. When d raises an integer grading N by one,

$$U_\theta = e^{i\theta N}, \quad B_{e^{i\theta}} = U_\theta B_1 U_\theta^\dagger.$$

If states and observables are transported as well, these are unitarily equivalent descriptions. A physical relative phase requires fixed comparison data. Squaring alone selects neither that data nor an arrow of time.

5.3 Curved squares can see phase without identifying it

For an arbitrary finite operator d , put $\Delta_0 = dd^\dagger + d^\dagger d$. The same expansion gives

$$B_\zeta^2 = \Delta_0 + \zeta^2 d^2 + \bar{\zeta}^2 (d^\dagger)^2.$$

The family is independent of *every* unit phase exactly when $d^2 = 0$. Its two nonconstant Fourier coefficients are d^2 and $(d^\dagger)^2$; both must vanish for constancy. But the sign pair $\zeta, -\zeta$ always gives the same square [23].

There can be further ambiguities. For

$$d = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \quad d^2 = 2I, \quad \Delta_0 = 5I,$$

one obtains $B_{e^{i\theta}}^2 = (5 + 4 \cos 2\theta)I$. The squares at $\theta = 0$ and $\pi/2$ are $9I$ and I , but θ and $-\theta$ give the same square and need not differ by a sign phase. Phase dependence is therefore not unique recovery even modulo sign.

For the graded Maurer–Cartan source the related identity is

$$d_a = d + [a, -], \quad d_a^2 = [F(a), -], \quad F(a) = da + a^2,$$

with graded brackets [10]. Here the operator curvature is the *action* of F , not necessarily the element F itself. Central curvature can be nonzero while its adjoint action vanishes. Ordinary cohomology is unavailable when $d_a^2 \neq 0$, but nonzero curvature does not forbid it in a representation where that curvature acts trivially. The curved-square correction is also not automatically the residual-Hessian correction of [theorem 2.1](#); relating them requires their common source, pairing and admissible variations.

5.4 Jordan and Lie readouts lose different information

In a finite unital matrix star algebra with positive cyclic trace τ , take $F = F^\dagger$ and define

$$C_F(X) = FX + XF, \quad A_F(X) = FX - XF.$$

Together they reconstruct $FX = (C_F X + A_F X)/2$ and $XF = (C_F X - A_F X)/2$. The Jordan map C_F is self-adjoint on the real Hermitian slice. The commutator is a derivation; $-iA_F$, not A_F itself, maps Hermitian observables to Hermitian observables and generates their unitary conjugation [20].

For a residual whose second derivative is $XY + YX$, the trace identity

$$\tau(F(XY + YX)) = \tau((FX + XF)Y)$$

identifies the ambient residual-Hessian correction with C_F on this real slice. For instance $F = \text{diag}(2, 1)$ and $X = E_{12} + E_{21}$ give $C_F X = 3X$, $A_F X = E_{12} - E_{21}$ and a Jordan quadratic pairing of six. This finite trace model does not automatically supply the graded signs, positive real slice or cyclic pairing of a physical q79 theory. A gauge row, a curved admissible stratum or additional physical residuals can also change its Hessian.

The information loss is exact in $\mathcal{A} = \bigoplus_{\alpha=1}^r M_{n_\alpha}(\mathbb{C})$. Write

$$F = F_0 + z, \quad z = \bigoplus_{\alpha} \lambda_{\alpha} I_{n_{\alpha}}, \quad \lambda_{\alpha} = \frac{\text{Tr } F_{\alpha}}{n_{\alpha}},$$

where each block of F_0 is traceless. The split is orthogonal for a positive weighted block trace. Knowing $[F, X]$ for *all* $X \in \mathcal{A}$ determines F only modulo this center, whereas $F = \frac{1}{2} C_F(1)$ recovers it from the complete Jordan map [18].

This last qualification is essential. A Hessian known only on an admissible tangent space may not include the unit or determine $C_F(1)$; restricted commutator observations can likewise leave more ambiguity than the center. Complete ambient readouts cannot be assumed from a few projected rows.

An exact group action permuting isomorphic blocks leaves one invariant real central coordinate per block orbit. For $M_2 \oplus M_2$, the two center coordinates reduce to one under an imposed block exchange. That counts the remaining *central ambiguity*, not all source parameters. The algebra, symmetry, trace, admissible variations and traceless data remain inputs until selected. A one-dimensional center of M_{27} is not a theorem that the physical 27-carrier model has one free constant, or that the unresolved scalar rows are central curvature.

5.5 Admissible geometry comes before physical stability

For an odd self-adjoint finite charge Q , consider the specific homogeneous cost

$$F = Q^2, \quad E(Q) = \frac{1}{2} \|Q^2\|^2, \quad L_Q(a) = Qa + aQ.$$

With a fixed positive trace pairing, $dE_Q(a) = \langle F, L_Q(a) \rangle$. If the admissible tangent space V contains the radial direction Q , stationarity would require

$$0 = dE_Q(Q) = 2 \|F\|^2,$$

so nonzero curvature cannot be stationary in that radial direction [16]. This is a no-go for this homogeneous cost and variation space, not for every nonlinear defect. Fixing a differential's principal symbol, a normalization stratum or boundary data may remove the radial variation, but that restriction is source data and must be justified.

On a smooth admissible submanifold S , tangent restriction of the ambient Hessian alone is insufficient. With Π its normal second fundamental form,

$$\text{Hess}_S(E|_S)(a, b) = \text{Hess}_{\text{amb}} E(a, b) + \langle \text{grad}_{\text{amb}} E, \Pi(a, b) \rangle.$$

This is the same covariant chain rule as [theorem 2.1](#), now applied to the constrained source geometry. The extra term may remain nonzero at a critical point of $E|_S$, because that point need not be ambient-critical.

The finite circle witness takes $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and

$$Q_0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \quad Q_1 = \begin{pmatrix} 0 & J \\ -J & 0 \end{pmatrix}.$$

Both are self-adjoint and odd for grading $\text{diag}(I_2, -I_2)$; $Q_0^2 = Q_1^2 = I_4$ and $Q_0Q_1 + Q_1Q_0 = 0$. Thus $Q_\theta = \cos \theta Q_0 + \sin \theta Q_1$ has constant square I_4 and cost two under the ordinary trace. At zero, the ambient Hessian along Q_1 is eight, while the normal-acceleration term is minus eight. The intrinsic circle Hessian is zero.

The circle therefore carries a protected first-order coordinate, not a damped mode proved by its positive ambient entry. This also does not select its traversal speed, identify it with the physical shared line, or turn its parameter into Lorentzian time. The useful synthesis is a source with an operator, metric, admissible variation geometry and shared representation, from which compatible readouts can follow. None of those source data can be reconstructed merely by renaming their projections.

6 Hodge, Green, and kernel shadows

6.1 The total-space Hodge package

Choose a smooth Hermitian metric on $E^\bullet$, a Hermitian metric on the compact boundaryless base J , and the associated density. Define

$$B_E = \overline{E} + \overline{E}^\dagger, \quad \Delta_E = B_E^2. \quad (6.1)$$

Its principal symbol is the Dolbeault symbol. Standard elliptic theory then gives a self-adjoint closure with compact resolvent [1].

Theorem 6.1 (Hodge and kernel package). *Under the preceding compactness, ellipticity, metric, and domain hypotheses, there exist:*

1. a finite-dimensional harmonic space $\ker \Delta_E$ and orthogonal projector P_0 ;
2. a smoothing heat semigroup $e^{-s\Delta_E}$ for $s > 0$;
3. a reduced Green operator

$$G_E = (\Delta_E|_{\ker \Delta_E^\perp})^{-1}(I - P_0) = \int_0^\infty (e^{-s\Delta_E} - P_0) \, ds; \quad (6.2)$$

4. finite-rank spectral projectors $\chi_I(\Delta_E)$ for bounded spectral intervals I ; and
5. local Schwartz kernels for the smoothing functional calculus.

Every item is determined by the same selected Δ_E .

The word “shadow” is literal but not mystical. The spectral resolution is the wave-like normal-mode description. The local kernel

$$K_s(x, y) = \langle x | e^{-s\Delta_E} | y \rangle$$

is the localized propagation-of-repair description. They are two representations of one operator. Neither representation, by itself, makes the semigroup a physical quantum evolution.

6.2 The wave-particle operator obtains a source candidate

The conditional wave-particle paper begins with a supplied positive operator A and forms

$$B_{\text{adm}} = P\chi(A)e^{-\tau_{\text{adm}}A}\chi(A)P[22].$$

At the present structural tier, [theorems 4.2](#) and [6.1](#) provide the exact candidate

$$A = \Delta_1 = D\Phi_{\text{MC}}(0)^\dagger D\Phi_{\text{MC}}(0) \quad (6.3)$$

on the cohesive benchmark. This closes the formerly missing mathematical arrow from nonlinear repair to the positive filtered operator on that benchmark.

It does not close the physical common-source theorem of the wave–particle paper. That stronger theorem must show that the state, Hamiltonian or generator, effects, instrument, damping channel, and selected MTT source all belong to one physical construction. Equation (6.3) supplies the positive repair operator, not the rest of quantum mechanics.

6.3 A stratified warning

The existence of a regular total-space operator does not imply a uniform family of fiberwise inverses. The following exact example is the useful model.

Proposition 6.2 (Regular complex, divergent fiberwise Green operator). *Let*

$$d_r = \begin{bmatrix} \text{diag}(1, 0, 0) & rI_3 \end{bmatrix} : \mathbb{C}^6 \longrightarrow \mathbb{C}^3$$

and let

$$B_r = \begin{bmatrix} 0 & d_r^\dagger \\ d_r & 0 \end{bmatrix}, \quad \Delta_r = B_r^2.$$

The entries of B_r and Δ_r are polynomial in r and remain finite at $r = 0$. For $r \neq 0$,

$$\text{spec}(\Delta_r) = \{0^{(3)}, (1 + r^2)^{(2)}, (r^2)^{(4)}\}.$$

At $r = 0$,

$$\text{spec}(\Delta_0) = \{0^{(7)}, 1^{(2)}\}.$$

Hence the smallest positive fiberwise eigenvalue is r^2 and the reduced fiberwise Green norm grows as r^{-2} .

Proof. The three rows of d_r are orthogonal with squared norms $1 + r^2, r^2, r^2$. These are the nonzero eigenvalues of both $d_r d_r^\dagger$ and $d_r^\dagger d_r$, with each nonzero singular value appearing in both blocks of Δ_r . The stated multiplicities and inverse norm follow directly. $\square$

At $r = 2$, the spectrum is $\{0^{(3)}, 4^{(4)}, 5^{(2)}\}$; at $r = 0$, the kernel jumps from dimension three to seven. The full complex is regular while the open-stratum contraction using $1/r$ is not. This is why the cohesive object cannot be replaced globally by its rank-three open-stratum cohomology.

7 Descent, holonomy, and compatible projection

7.1 Nonlinear Cech descent

Let $\{\mathcal{M}_i\}$ be local Riemannian configuration charts with local defect spaces $\mathcal{E}_i$. Suppose the overlap maps

$$U_{ij} : \mathcal{M}_i \rightarrow \mathcal{M}_j, \quad V_{ij} : \mathcal{E}_i \rightarrow \mathcal{E}_j$$

are respectively Riemannian isometries and unitary maps, and

$$\Phi_j \circ U_{ij} = V_{ij} \circ \Phi_i. \tag{7.1}$$

Theorem 7.1 (Nonlinear repair descent). *The local costs $\mathcal{C}_i = \frac{1}{2} \|\Phi_i\|^2$, negative-gradient fields $X_i = -\text{grad } \mathcal{C}_i$, and their local flows $R_i(s)$ obey*

$$\mathcal{C}_j \circ U_{ij} = \mathcal{C}_i, \quad (7.2)$$

$$TU_{ij}X_i = X_j \circ U_{ij}, \quad (7.3)$$

$$U_{ij} \circ R_i(s) = R_j(s) \circ U_{ij}. \quad (7.4)$$

Therefore the nonlinear Čech equalizer

$$\Gamma(\mathcal{M}) = \{(x_i) : x_j = U_{ij}x_i \text{ on every overlap}\}$$

is preserved. Compatible fixed points, basins, and existing asymptotic limits descend. A single-valued global basin label must lie in

$$\bigcap_{\gamma} \text{Fix}(\text{Hol}_{\gamma})$$

for the induced label local system.

Proof. Equation (7.1) and unitarity give equality of the local costs. Differentiating that equality and using that U_{ij} is an isometry intertwines the gradient fields. Uniqueness for the local initial-value problem then intertwines the flows. Passing to an existing limit commutes with the continuous overlap maps. Transport around a cycle gives the holonomy-fixed condition on a global label. $\square$

This theorem does not select one basin. It says exactly when local repair is one global process. Unique actualization requires an additional theorem that one allowed global basin is selected, or an explicit probability law when several remain. Measurement need not be a privileged primitive here; it is one physical process capable of changing the local defect map and its basins.

7.2 Tangent and spectral projection

For a finite connected cover with unitary local-system transitions, the linear equalizer is concretely

$$\Gamma(\mathcal{H}) = \ker d_0, \quad (d_0\psi)_{ij} = \psi_j - U_{ij}\psi_i.$$

Transport along a spanning tree determines all components from one root vector; the remaining edges impose the cycle-holonomy fixed equations. Thus the global space is the intersection of those fixed spaces, not a tensor product over overlapping charts. Projectors descend only when $U_{ij}P_i = P_jU_{ij}$. These are the finite linear conclusions of the local Čech–Hilbert record cited in the evidence section.

Differentiate (7.1) at a compatible zero-defect fixed point. The Jacobians and Hessians intertwine, so the tangent heat semigroup preserves the linear Čech equalizer. The Hilbert-space description is thus the linearization of nonlinear descent, not its replacement.

Theorem 7.2 (Compatible projection of the repair calculus). *Let $H \geq 0$ be a self-adjoint repair Hessian on a Hilbert space and let P be an orthogonal projector reducing H . Put $H_P = H|_{\text{Ran } P}$. Then, on their declared domains,*

$$Pe^{-sH} = e^{-sH_P}P, \quad P\chi_I(H) = \chi_I(H_P)P, \quad PG = G_PP, \quad (7.5)$$

where the Green identity is taken on the corresponding harmonic complements. A unitary equivalence T of configuration spaces and a unitary equivalence U of residual spaces, satisfying $\Phi'(Ta) = U\Phi(a)$, transport the nonlinear cost jet and the tangent functional calculus at compatible zero-defect backgrounds.

For an isometric embedding rather than an onto equivalence, the cost jet is transported only as a pullback. Full tangent transport additionally requires, on the common operator domains,

$$J'T = UJ, \quad J'^{\dagger}U = TJ^{\dagger}.$$

These identities make the embedded tangent range reducing for the self-adjoint Hessian. Without the second identity only $T^{\dagger}H'T = H$ is implied, not $H'T = TH$.

Proof. The first three identities are consequences of the spectral theorem for a reducing subspace. The residual identity gives the pullback equality $\mathcal{C}'(Ta) = \mathcal{C}(a)$ and hence equality of all pulled-back cost derivatives. At a zero-defect background the two displayed derivative identities give $H'T = J'^{\dagger}J'T = J'^{\dagger}UJ = TJ^{\dagger}J = TH$. For onto unitary maps the adjoint identity follows from the chain identity. For an embedding it is an extra hypothesis. The common-domain self-adjoint intertwining then gives the claimed functional calculus. $\square$

For example, let $\Phi(x) = x$, $\Phi'(x, y) = x + y$, $Tx = (x, 0)$, and $U = 1$. The residual and cost pull back exactly, but

$$H = 1, \quad H' = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad H'Tx = (x, x) \neq (x, 0) = THx.$$

The transverse response prevents reduction. This explains why a dimension match, an isometric compression or a compatible cost restricted to a subspace cannot replace a full operator intertwiner.

Retained projectors are not the whole ambient projector. There is a second, independent completeness requirement in the older continuum-to-finite cutset record [31]. Even exact reducing intertwining gives $\Pi'T = T\Pi$, not necessarily $\Pi' = T\Pi T^{\dagger}$. For a concrete counterexample, take

$$H = \text{diag}(0, 1), \quad H' = \text{diag}(0, 1, 0), \quad T(x, y) = (x, y, 0).$$

The intertwining residual is zero, both positive spectral gaps are one, and the range of T is reducing. Nevertheless,

$$\Pi' - T\Pi T^{\dagger} = \text{diag}(0, 0, 1), \quad \|\Pi' - T\Pi T^{\dagger}\| = 1.$$

Thus the packet's unrestricted whole-projector estimate $\|\Pi' - T\Pi T^{\dagger}\| \leq 2\epsilon_H/g$, based only on $\epsilon_H = \|H'T - TH\|$ and a gap g , is false. It must not be used to promote the ambient kernel. In the exact reducing case the precise identity is

$$\Pi' = T\Pi T^{\dagger} + \Pi'_{\perp},$$

where $\Pi'_{\perp}$ projects onto the zero modes in $(\text{Ran}T)^{\perp}$. Equality of the whole projectors requires $\Pi'_{\perp} = 0$; the retained intertwining identity needs no such assumption.

For approximate comparisons, the spectral object must also be fixed correctly. A small positive perturbation of a zero eigenvalue changes its literal kernel, although an isolated

spectral cluster can persist. On a common Hilbert space let A, B be self-adjoint, let $A - B$ extend to a bounded operator on their common operator domain, and let a common rectifiable resolvent contour Γ isolate the two clusters being compared. Writing $M_A = \sup_{z \in \Gamma} \|(z - A)^{-1}\|$, and similarly M_B , the resolvent identity and contour integral give

$$\|\Pi_A - \Pi_B\| \leq \frac{\text{length}(\Gamma)}{2\pi} M_A M_B \|A - B\|.$$

Indeed, subtract the two Riesz integrals and insert $(z - A)^{-1} - (z - B)^{-1} = (z - A)^{-1}(A - B)(z - B)^{-1}$. This is a cluster-comparison estimate, not a bound derived from an embedding residual alone. Identifying either cluster with the full physical harmonic sector requires a separate completeness certificate.

For a non-isometric Fourier–Mukai or BHT transform, derived equivalence alone is insufficient. It may preserve Ext groups, Yoneda products, and the formal Maurer–Cartan deformation problem after dg enhancement [13, 48, 5]. It does not automatically preserve the Hermitian metric, adjoint, Hodge spectrum, or numerical kernel. Those require a transported metric theorem or a quantified comparison defect.

The transported-metric result makes the latter task concrete. For a bounded invertible chain/product map T , with domains also carried by T , set

$$g_{\text{tr}} = (T^{-1})^\dagger g T^{-1}.$$

Then T is unitary for this transported metric. If a separately selected physical metric is $g_{\text{phys}}(x, y) = g_{\text{tr}}(x, Ey)$, its positive distortion E changes the degree- k adjoint by

$$D_k^{\dagger \text{phys}} = E_k^{-1} D_k^{\dagger \text{tr}} E_{k+1}, \quad C_k = E_{k+1} D_k - D_k E_k.$$

Thus $C_k = 0$ is the metric-chain compatibility test. The exact witness in the transported-metric record has transported Hodge matrix I_2 but physical matrix $\text{diag}(4, 1/9)$. Polar normalization does not automatically repair the product: if $T = US$, its positive factor must itself respect the differential and product for U to be a dg-algebra map. These are explicit comparison tests, not a construction of the physical BHT operator or its HYM metric.

8 What survives a projection: compression, elimination and transfer

A retained space can inherit an operator without inheriting its entire algebra. Agreement of quadratic forms, products, Green operators and interaction vertices are different requirements. The following comparison integrates the finite compression and transfer results without identifying their carriers with a physical mass matrix. The general transfer machinery is standard homological algebra [14, 15]; the source-specific content is the exact finite realization and its constraints.

8.1 Leakage measures the failure of an algebra to descend

Let P be an orthogonal projector, $Q = I - P$, and write $\Phi_P(S) = PSP|_{P\mathcal{H}}$. For bounded operators,

$$\Phi_P(S)\Phi_P(T) - \Phi_P(ST) = -PSQTP.$$

Insert $I = P + Q$ between S and T to prove the identity. The missing term is an excursion out of the retained space and back. Compression is multiplicative on a chosen algebra only when its excursions vanish, for example when the space reduces that algebra.

For commuting self-adjoint A, B , put $L_A = QAP$ and $L_B = QBP$. Then

$$[\Phi_P(A), \Phi_P(B)] = L_B^\dagger L_A - L_A^\dagger L_B, \quad \|[\Phi_P(A), \Phi_P(B)]\| \leq 2 \|L_A\| \|L_B\|.$$

The exact three-dimensional witness takes $Q = \frac{1}{3}\mathbf{1}\mathbf{1}^\top$, $A = \text{diag}(1, 0, 0)$, $B = \text{diag}(0, 1, 0)$. Its compressed commutator, embedded in the ambient space, is

$$\frac{1}{9} \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}.$$

Commuting observables can therefore acquire a noncommutative compression. This does not prove that every projection does so, or derive canonical commutation relations, a Born state or Planck's constant [21].

The gauge-descent example supplies a concrete origin for this projector. On $\mathbb{R}^3$, set $H_0 = 2P + 5Q$ and $F(u) = H_0 u + (u^\top u)u$, with repair equation $\dot{u} = -F(u)$. The inner product with u shows that $F(u) = 0$ only at zero. The tangent operator is H_0 ; its isolated eigenvalue-two space selects P . Signed permutations preserve the construction. Their conjugation action on the retained full matrix algebra has kernel $\{\pm I\}$, leaving a faithful quotient of order six. In general one must use the stabilizer of the chosen fixed point and an isolated spectral subspace. This is a descent mechanism, not selection of the Standard Model gauge group [34].

8.2 The internal operator cannot be omitted

For $R = QRQ$, write

$$E_R(S, T) = PS RTP, \quad E_R(S, T) - E_Q(S, T) = PS(R - Q)TP.$$

Raw compression uses $R = Q$. Feshbach elimination uses the inverse of $Q(H - z)Q$ on $Q\mathcal{H}$, when it exists, and gives

$$P(H - z)P - PHQ [Q(H - z)Q]^{-1}QHP.$$

For P the first two coordinates of

$$H = \begin{pmatrix} 1 & 0 & 1/2 & 0 \\ 0 & 2 & 0 & 0 \\ 1/2 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix},$$

the raw $PHQHP$ is $\text{diag}(1/4, 0)$, whereas the inverse-weighted term at zero is $\text{diag}(1/12, 0)$. The effective operator is $\text{diag}(11/12, 2)$.

Homotopy transfer inserts a different object again: $h = d^\dagger G$, of degree minus one, rather than a degree-zero projector or resolvent. Its ternary operation has three inputs and two signed tree terms. An exterior differential algebra makes the distinction tangible: take $da = db = 0$, $dc = ab$, an orthonormal monomial basis and harmonic complement spanned by ab, c . Here $G = Q$, but $h(ab) = c$; the source sign convention gives $m_3(a, a, b) = ac$. Equality of G and Q has not removed the differential or changed a ternary map into a binary compression defect. This Heisenberg differential-algebra example is an auxiliary witness, not the global q79 topology [19].

8.3 Covariance completion changes the harmonic content

Start with $A = M_3(\mathbb{Q}(\omega))$, $\omega^2 + \omega + 1 = 0$, a qutrit Weyl pair $ZX = \omega XZ$, and commuting automorphisms $\alpha = \text{Ad } X$, $\beta = \text{Ad } Z$. The two-direction crossed exterior calculus has

$$K_W = A \oplus (A\theta_x \oplus A\theta_z) \oplus A\theta_x\theta_z, \quad \dim K_W = (9, 18, 9).$$

Its relations are $\theta_x a = \alpha(a)\theta_x$, $\theta_z a = \beta(a)\theta_z$, with anticommuting square-zero forms, and $da = (\alpha(a) - a)\theta_x + (\beta(a) - a)\theta_z$. The finite metric gives Weyl-mode Laplacian eigenvalues 0, 3, 6, with multiplicities 1, 4, 4 before tensoring by forms. Harmonic dimensions are (1, 2, 1). Their scalar-coefficient representatives form an exterior subalgebra, so transfer to these harmonics has $m_n = 0$ for $n \geq 3$. A Hodge package alone does not manufacture nonzero interactions [36].

On harmonic degree one reflection acts as $-I_2$, and the quarter-turn as $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. The associated determinant-twisted rank-six local-system construction has a rank-two Reynolds sector and four complementary strain directions. But the selected full-chain reflection and quarter-turn fail to preserve the old product. Cohomological compatibility is not full differential graded algebra (DGA) compatibility [32].

Closing the direction orbit under inversion and the quarter-turn gives $+x, -x, +z, -z$, hence a symmetric DGA A_{sym} with degree dimensions (9, 36, 54, 36, 9), totaling 144. Squared norm 1/2 per signed edge retains the eigenvalues 0, 3, 6. The finite order-36 group $(\mathbb{Z}_3 \times \mathbb{Z}_3) \rtimes C_4$ now acts by DGA automorphisms. The old complex embeds isometrically as a reducing cochain summand, but not as a product subalgebra; its orthogonally compressed product is nonassociative off harmonics [42].

The completion has harmonic dimensions (1, 4, 6, 4, 1), not (1, 2, 1). The twelve extra classes exemplify the ambient modes that a retained intertwiner cannot exclude. These complexes are not quasi-isomorphic. Their dimensions are neither spacetime dimension, the 27-dimensional finite carrier, nor the rank-102 continuum response bundle.

8.4 A first-jet quotient is not a deletion theorem

Let O span the odd differences $o_i = \theta_{+i} - \theta_{-i}$, and E the even sums $e_i = \theta_{+i} + \theta_{-i}$, for $i = x, z$. Formal connection transport along one direction gives

$$\frac{(e^{h\nabla_i} - 1)\theta_{+i} + (e^{-h\nabla_i} - 1)\theta_{-i}}{h} = \frac{\sinh(h\nabla_i)}{h}o_i + \frac{\cosh(h\nabla_i) - 1}{h}e_i.$$

The first jet is $\nabla_i o_i$; the even channel starts with $h\nabla_i^2 e_i/2$. Reversal selects O as the universal first-symbol landing plane, with unique orthogonal projector $(I - f^2)/2$ for the quarter-turn f . This formal identity does not construct a small-spacing family for the finite qutrit translations [37].

The exact harmonic quotient is

$$\mathcal{J} = \langle E \rangle \subset \Lambda(O \oplus E), \quad \Lambda(O \oplus E)/\mathcal{J} \simeq \Lambda(O).$$

The ideal has dimensions (0, 2, 5, 4, 1), totaling twelve. This is a strict associative quotient of the harmonic algebra, not the nonassociative compression of the old full cochain complex.

Scaling $o_i \mapsto o_i$, $e_i \mapsto h e_i$ defines a family of algebra maps. At nonzero scalar h it is invertible; at zero its image has dimension four and kernel $\mathcal{J}$. This statement about specialized images is not a flat family whose fiber dimension drops from sixteen to four. Specialization of the image module over $\mathbb{Q}(\omega)[h]$ need not equal the image of the specialized map. The quotient and suppression order are exact; a flat Rees degeneration requires a separate construction. Higher-jet data are not thereby physically absent modes.

8.5 The response retract keeps the missing classes

An actual strong deformation retract keeps these classes instead of discarding cohomology. Its target is

$$T = K_W \oplus \mathcal{J}, \quad \dim T = (9, 20, 14, 4, 1), \quad \dim T = 48.$$

For inclusion i , projection p and degree-minus-one homotopy h , the certified identities are

$$pi = I_T, \quad dh + hd = I - ip, \quad h^2 = ph = hi = 0.$$

The target differential $m_1 = d_T$ is not zero. This is a response model, not a minimal model consisting only of cohomology. Transfer gives $m_2 = p\mu(i \otimes i)$ and, in the source convention,

$$m_3(a, b, c) = p\mu(h\mu(ia, ib), ic) \\ - (-1)^{|a|} p\mu(ia, h\mu(ib, ic)).$$

The arity-three identity expresses the associator of m_2 as the corresponding differential of m_3 : strict associativity has a controlled chain-level replacement [40].

The frozen computation evaluates every target basis pair and triple and then all degree-admissible quadruples. Both m_3, m_4 are nonzero; the arity-three and arity-four identities pass exactly. The quadruple audit checks 3,869,500 degree-admissible inputs, with 693,208 nonzero m_4 values [41]. These archived computations are not rerun here. The nonzero inputs use response modes, so they do not contradict the vanishing of higher products on the harmonic subalgebra.

There is also an all-arity proof. In the fixed source basis take degree-one x, y, w and degree-zero z , labeled $x = C_{0,0,1}$, $y = C_{1,0,1}$, $z = C_{1,0,0}$, $w = C_{2,0,1}$. The final index is an exterior-basis mask, not a third Weyl charge. A two-parity state recurrence proves

$$m_{2r+3}(x^{\otimes(2r+1)}, y, z) = \frac{2 + \omega}{4 \cdot 12^r} w, \\ m_{2r+4}(x^{\otimes(2r+2)}, y, z) = -\frac{\omega}{8 \cdot 12^r} w \quad (r \geq 0).$$

The hierarchy therefore does not truncate. A separate invariant-subspace argument gives $m_n = 0$ for $n \geq 3$ when at least $n - 1$ inputs lie in $\mathcal{J}$ [26]. These are not exhaustive tables: the cited snapshot does not compute the complete m_5 table or enumerate every arity-five Stasheff residual. The transfer theorem supplies formal identities; exhaustive table verification is a distinct implementation check.

8.6 One source map can transport the whole hierarchy

Let $\varphi : A \rightarrow A'$ be a degree-zero DGA morphism and $\psi : T \rightarrow T'$ a retained chain map between normalized contractions. If

$$\varphi i = i' \psi, \quad p' \varphi = \psi p, \quad \varphi h = h' \varphi,$$

then $\psi m_n = m'_n \psi^{\otimes n}$ for every $n \geq 1$. Move φ through each decorated transfer tree: its product identity handles vertices, the homotopy square handles internal edges, and the other squares handle leaves and root. Degrees, hence signs, are unchanged. This proves all-arity naturality rather than extrapolating from low-order tests [17].

The finite translation and Fourier generators preserve the complete 144-to-48 contraction. Their order-36 covariance therefore acts on every transferred operation. At the continuum

level a same-source product-preserving map, reducing domains, metric and physical symmetry lift must still be supplied. Approximate squares require defect bounds; they do not give exact naturality.

The finite program thus has an exact response hierarchy and a finite set of sufficient source identities for transporting it. It has not selected the continuum HYM source or its Lorentzian/cyclic action. These m_n are not measured Yukawa or gauge couplings: they specify what must be preserved before that physical interpretation is justified.

9 Nested repair and curvature-preserving transfer

There are two different places where a differential can act: inside a configuration, and on the space of configurations. Keeping them distinct allows a curved internal charge to coexist with an ordinary outer complex. The six source records discussed here join that observation to the Heisenberg example in [section 8.2](#), then pass from central curvature to a positive noncentral coefficient algebra. They are finite or local structural results, not a new physical endpoint.

9.1 Inner curvature and an outer nilpotent differential

Let M be a smooth Riemannian configuration manifold with a smooth internal charge Q_x , defect $\Phi(x) = Q_x^2$, and a declared positive fiber metric. Put $V = \frac{1}{2} \|\Phi\|^2$, allowing an explicitly normalized trace in the finite examples below. On forms over M , define

$$d_{\beta,V} = e^{-\beta V} d_M e^{\beta V} = d_M + \beta dV \wedge, \quad \beta \geq 0.$$

This is Witten deformation [\[51\]](#). Conjugation gives $d_{\beta,V}^2 = 0$ for every smooth V , whether or not the internal defect vanishes [\[27\]](#). Thus $[Q_x, -]^2 = [\Phi(x), -]$ can be nonzero while the outer differential is nilpotent. These operators act on different spaces; neither identity repairs the other by changing its meaning.

For a closed compact M , the usual elliptic realization gives the positive operator

$$H_{\beta,V} = d_{\beta,V} d_{\beta,V}^\dagger + d_{\beta,V}^\dagger d_{\beta,V}$$

and its contractive heat semigroup. Boundary or noncompact settings need compatible adjoints and self-adjoint domains; the polynomial computations below are formal differential-operator identities, not domain certificates. On the configuration line let $A_V = \partial_x + \beta V'$. Then

$$\begin{aligned} H_0 &= A_V^\dagger A_V = -\partial_x^2 + \beta^2 (V')^2 - \beta V'', \\ H_1 &= A_V A_V^\dagger = -\partial_x^2 + \beta^2 (V')^2 + \beta V''. \end{aligned}$$

At a critical point, V'' contains the full residual-Hessian term from [theorem 2.1](#), not just the Gram term. For $\Phi(x) = 1 + x^2$, one has $V = \frac{1}{2}(1 + x^2)^2$ and, at zero,

$$\Phi = 1, \quad \Phi' = 0, \quad V' = 0, \quad |\Phi'|^2 = 0, \quad \Phi\Phi'' = 2, \quad V'' = 2.$$

The opposite potential terms -2β and 2β are not opposite spectral-positivity claims: positivity belongs to the factorized differential operators, including their kinetic terms.

This construction retains the defect's full second jet, but it does not make the Witten operator equal to the internal curvature or to the finite-dimensional Hessian matrix. On a closed compact M , multiplication by $e^{\beta V}$ identifies its cohomology with de Rham cohomology. Nilpotence alone therefore does not select zero defect or a unique physical configuration. The supplied β is a localization parameter, not Lorentzian time or a derived thermodynamic temperature.

9.2 What the eliminated pair leaves behind

For the auxiliary algebra $\Lambda(a, b, c)$ with $dc = ab$, the six harmonic representatives are

$$H = \text{span}(1, a, b, ac, bc, abc).$$

Let i include them, let p remove the pair c, ab , and set $h(ab) = c$, with $h = 0$ otherwise. Then

$$pi = I_H, \quad dh + hd = I - ip, \quad h^2 = ph = hi = 0.$$

Under the transfer convention already used in [section 8.5](#), the two nonzero examples are

$$m_3(a, a, b) = ac, \quad m_3(a, b, b) = -bc \quad [25].$$

They remember the removed relation $dc = ab$; a cohomology-dimension count does not encode them. Minimal-model and strict-unital constructions are standard homological algebra [\[12, 6\]](#).

With weights $w(a) = w(b) = 1$, $w(c) = 2$, every source operation and contraction map preserves total integer weight, hence also its reduction modulo three. This is a representation-level phase statement, not a construction of the continuum shared line or an identification of Heisenberg/Nil with the global q_79 manifold.

The top-form functional $\tau_A(u) = [abc]u$ is graded cyclic but is not a positive Hilbert norm. It also carries weight four: under the stated circle action, $\tau_A(g_\theta u) = e^{4i\theta} \tau_A(u)$. Thus its pairing is covariant with that character, not automatically an invariant scalar pairing. An invariant scalar interpretation needs the inverse-character line or equivalent compensating data. Preserving a cyclic contraction and preserving weights are compatible assertions; neither supplies that additional scalar trivialization.

9.3 Which data a nested contraction must preserve

A quasi-isomorphism is insufficient for this purpose. The restricted transfer record requires the same admissible base, typing and boundary data; maps p, i, h satisfying the contraction and side identities; first-order and adjoint intertwiners; compatible cyclic and Hilbert pairings; a unit and equal normalized repair potentials; and shared-line equivariance [\[35\]](#). The maps must be parallel over the configuration base, with a common β , to intertwine the outer Witten operators. Transfer trees also need a terminating filtration or a proved convergence domain.

For two algebraically compatible contractions the composite is

$$p = p_2 p_1, \quad i = i_1 i_2, \quad h = h_1 + i_1 h_2 p_1.$$

These formulas preserve the normalized contraction identities. Analytic admissibility of the composite still needs a common transfer-control class; convergence on unrelated domains is not automatically enough. Likewise, local arrows alone do not prove effective overlap descent.

Here is the same-source central example. Let Γ_A be form parity, take $S = \mathbb{R}^{2|2}$, and define

$$R(x) = \begin{pmatrix} 1 & x \\ -x & 1 \end{pmatrix}, \quad C(x) = \begin{pmatrix} 0 & R(x) \\ R(x)^\top & 0 \end{pmatrix}, \quad Q_A = d \otimes I_4 + \Gamma_A \otimes C(x).$$

Since $C(x)^2 = (1 + x^2)I_4$ and $d\Gamma_A = -\Gamma_A d$,

$$Q_A^2 = (1 + x^2)I_{32}, \quad Q_H = \Gamma_H \otimes C(x), \quad Q_H^2 = (1 + x^2)I_{24}.$$

Tensor the preceding contraction with I_4 . It intertwines Q , its declared adjoint and curvature. Rank-normalized trace gives $V = \frac{1}{2}(1 + x^2)^2$ on both fibers; raw traces would differ with

rank. Because p is independent of x , it also intertwines the formal outer Witten blocks. The same p, i, h produce the nonzero m_3 .

Strict intertwining and transfer remain different claims: the source DGA has no ternary operation to project, while its minimal model has one. Equality of normalized costs is verified for this source, not inferred from any arbitrary rank-reducing map.

9.4 Two ways to retain noncentral curvature

The infinitesimal route works over $D = \mathbb{Q}[\epsilon]/(\epsilon^2)$, with matrix indices starting at zero and parities $(0, 1, 0)$. Put

$$q = E_{10} + \epsilon E_{21}, \quad F = q^2 = \epsilon E_{20}, \quad [F, E_{00}] = F \neq 0.$$

For homogeneous X , the graded commutator $d_q X = qX - (-1)^{|X|} Xq$ satisfies $d_q^2 X = [F, X]$ and $d_q F = 0$. Every two-curvature product FXF vanishes over D . For the Heisenberg contraction and $\delta = \Gamma_A \otimes q$,

$$(h\delta)^2 = (\delta h)^2 = 0, \quad (I - h\delta)^{-1} = I + h\delta.$$

This is exact algebraic termination, not a small numerical error [28]. The 24-to-18 module ranks are over D , hence double over $\mathbb{Q}$. The coefficient-of- ϵ supertrace is nondegenerate and graded cyclic, not positive. No nonzero nilpotent self-adjoint operator can represent this F on a positive finite Hilbert space.

The positive route instead takes $S = \mathbb{R}^{1|2}$ and

$$q(x) = \begin{pmatrix} 0 & 1 & x \\ 1 & 0 & 0 \\ x & 0 & 0 \end{pmatrix}, \quad F(x) = q(x)^2 = \begin{pmatrix} 1+x^2 & 0 & 0 \\ 0 & 1 & x \\ 0 & x & x^2 \end{pmatrix}.$$

Here $q = q^\top$, $F = q^\top q \geq 0$, and $[F(0), E_{12}] = E_{12}$. The same form contraction gives $Q_A = d \otimes I_3 + \Gamma_A \otimes q$ on 24 module directions and $Q_H = \Gamma_H \otimes q$ on 18. It retains $Q_A^2 = I_A \otimes F$, adjoint intertwining and the normalized cost

$$V(x) = \frac{1}{3}(1+x^2)^2, \quad V''(0) = \frac{1}{3} \|F'(0)\|_{\text{HS}}^2 + \frac{1}{3} \langle F(0), F''(0) \rangle_{\text{HS}} = \frac{2}{3} + \frac{2}{3} = \frac{4}{3} \quad [29].$$

The omitted half would be a genuine error in a Gram-only Hessian. The positive norm uses ordinary trace; graded cyclicity uses supertrace. They are separate pairings, not one automatically positive cyclic form.

A wording correction is necessary here. The self-adjoint coefficient matrix is q , not the full module superconnection:

$$Q_A - Q_A^\dagger = (d - d^\dagger) \otimes I_3 \neq 0.$$

Indeed $d(c) = ab$ while $d^\dagger(c) = 0$ in the monomial metric. The positive square $Q_A^2 = I_A \otimes F$ and all displayed intertwiners remain exact, but Q_A^2 must not be renamed $Q_A^\dagger Q_A$. Neither this module dimension nor the central example's outer-form dimension identifies the different CBF response carrier, the physical 27-carrier, or the rank-102 bundle.

9.5 Curvature as retained coefficients

The positive curvature is not nilpotent. Nevertheless the strict-unital extension record closes the local curvature-insertion problem for this particular tensor construction [39]. Keep

$B = \text{End}(\mathbb{R}^{1|2})[x]$, with $d_B = [q, -]$ graded and $d_B^2 = [F, -]$, as the coefficient algebra. Transfer only the Heisenberg form factor.

For this six-class minimal model, $m_n = 0$ for $n \geq 4$. A nonzero internal homotopy introduces c ; a later homotopy cannot remove it to recover ab , and two branches carrying c multiply to zero. Hence every tree with at least two homotopies vanishes. This is specific to the displayed exterior algebra and contraction, not a universal termination theorem.

Use total $\mathbb{Z}_2$ grading for the supermatrix extension; the integer form grading alone does not give an integer grading of the finite matrices. On $K = H \otimes B$, the source convention is

$$\begin{aligned} M_0 &= -1_H \otimes F, \\ M_1(u \otimes b) &= (-1)^{|u|} u \otimes d_B b, \\ M_n(u_1 \otimes b_1, \dots, u_n \otimes b_n) &= (-1)^\chi m_n(u_1, \dots, u_n) \otimes b_1 \cdots b_n \quad (n = 2, 3), \\ \chi &= \sum_{i < j} |b_i| |u_j|, \quad M_n = 0 \quad (n \geq 4). \end{aligned}$$

The ordered coefficient product is essential; B is not commutative. At fixed x , K has dimension $6 \cdot 9 = 54$, an operator algebra carrier rather than the 18-dimensional module above.

The curved arity-one identity reads

$$M_1^2(X) + M_2(M_0, X) - M_2(X, M_0) = 0.$$

Thus $M_0 = -F$ is a sign convention, not a claim that positive curvature became a negative repair norm. In every higher curvature insertion the H factor is the strict unit, so $M_n(\dots, M_0, \dots) = 0$ for $n \geq 3$. The remaining identities reduce to Bianchi, the graded derivation rule and the Heisenberg m_2/m_3 identities through arity five. No term remains at higher arity. The frozen record checks the 20,736 and 248,832 form/parity cases at arities four and five; those tables are not rerun here.

The scalar-curvature notation $m_0 = \lambda 1$ in the earlier finite example must likewise be interpreted in parity grading, or with a formal parameter of degree two if integer cohomological degrees are required. A real degree-zero scalar is not an integer-degree-two curvature merely because it is called even.

This closes the local algebraic conjunction of noncentral curvature, positive cost, outer nilpotent repair and transferred interactions. It does not require an infinite curvature-insertion sum in this model. There is no contradiction with the nontruncating CBF hierarchy of [section 8.5](#): that target is a different, nonminimal 48-dimensional response retract. A selected physical factorization, global descent, domains, shared differential line, signed action and normalization still require their own evidence. The examples explain what those data could organize; they do not select them.

10 Global descent and the actual rational topology

The local tensor examples answer how a coherent algebra can retain curvature under transfer. Two further records separate the global questions: how projective charts glue that algebra, and which rational form factor the q79 topology actually supplies. Neither question is answered by identifying every circle construction with the auxiliary Heisenberg example.

10.1 What descends through projective overlaps

Keep the projective atlas of [theorem 4.1](#). On a good cover, let G_{ij} be even unitary transitions satisfying $G_{ij}G_{jk}G_{ki} = \alpha_{ijk}I$, and suppose odd coefficient charges obey $q_i = G_{ij}q_jG_{ij}^{-1}$. Set

$$B_i = \text{End}(E_i), \quad F_i = q_i^2, \quad d_i = [q_i, -].$$

Conjugation is an honest algebra cocycle even though the module cocycle is projective. Squaring and taking graded commutators give

$$F_i = \text{Ad}(G_{ij})F_j, \quad d_i \text{Ad}(G_{ij}) = \text{Ad}(G_{ij})d_j.$$

Thus the earlier endomorphism untwisting extends to a curved differential, not just to the zero-curvature case [\[43\]](#). The coefficient charge need not commute with the transitions; it need only transform correctly.

Suppose the form-factor contractions have honest overlaps T_{ij} preserving the unit and intertwining their transferred m_n . Then

$$R_{ij} = T_{ij} \hat{\otimes} \text{Ad}(G_{ij})$$

has an honest triple law. For the coefficient operations of [section 9.5](#),

$$R_{ij}M_{n,j} = M_{n,i}R_{ij}^{\otimes n}, \quad n \geq 0.$$

For $n = 0$ this means transport of curvature. For $n \geq 2$ it follows from transport of m_n and the ordered coefficient product; even transitions preserve the Koszul signs. Hence the local curved identities glue without introducing another curvature-insertion series. This imports the overlap extension; it does not duplicate the untwisting proof.

An ordinary endomorphism sheaf is not necessarily a globally split matrix algebra. In the locally free case it can be an Azumaya algebra with nontrivial Brauer class [\[7\]](#). The scalar twist cancels from conjugation, not from the original module. For the same reason, central shared-line holonomy cannot be reconstructed from this adjoint action alone. The Hermitian differential line remains separate source data.

10.2 A noncommuting witness and its metric boundary

Let X, Z be the qutrit shift and clock, with $Xe_j = e_{j+1}$, $Ze_j = \omega^j e_j$, and $\omega^3 = 1$. On $\mathbb{C}^{1|2} \otimes \mathbb{C}^3$, take

$$G_{01} = I_3 \otimes X, \quad G_{12} = I_3 \otimes Z, \quad G_{20} = \omega(G_{01}G_{12})^{-1},$$

so their product is ωI_9 , while the adjoint triple is the identity. A nonidentity triple in this chosen atlas does not by itself certify that a global gerbe class is nonzero. The source charge is

$$q_0 = (E_{01} + E_{10}) \otimes I_3 + E_{02} \otimes X + E_{20} \otimes X^\dagger.$$

It is odd and self-adjoint, and a clock overlap moves it nontrivially. The transformed charges glue, their $F_i = q_i^\dagger q_i = q_i^2$ are positive and noncentral, and every chart has

$$V_i = \frac{1}{18} \text{Tr}(F_i^\dagger F_i) = \frac{4}{3}.$$

Here self-adjointness concerns the displayed coefficient q_i , not the full form-module operator corrected in [section 9.4](#). The witness tests actual conjugation, not a commuting-factor substitute.

There is also a useful limit to this example. Its Heisenberg overlap acts by ω^w , where the top form abc has weight four. Consequently its top-form functional satisfies

$$\tau_A(Ta Tb) = \omega \tau_A(ab),$$

not scalar invariance. The finite check certifies algebraic equivariance and the triple law, not all scalar cyclic-pairing hypotheses of the general descent theorem. A line-valued pairing, or the inverse-character line, supplies the needed compensation. This does not invalidate the algebraic descent or the positive coefficient trace.

Likewise, coordinate-dependent transitions require the connection law, not only matrix conjugation. In a local trivialization it has the form

$$\mathcal{A}_i = G_{ij} \mathcal{A}_j G_{ij}^{-1} - (dG_{ij}) G_{ij}^{-1},$$

up to central terms invisible on endomorphisms. Parallel overlap maps, compatible adjoints and common domains are needed before concluding that the corresponding Hodge and Witten operators glue. Unitary conjugation does preserve the normalized coefficient cost; it does not select those connection or physical-domain data.

10.3 The q79 circle extension, not a Nil replacement

Let B be the selected K3 surface, with classes

$$H^2 = 2, \quad \delta^2 = -4, \quad H \cdot \delta = 0,$$

and primitive δ . The topology under discussion is $X_{q79} = P_\delta \times S_{\text{shared}}^1$, where the principal circle bundle $P_\delta \rightarrow B$ has Euler class δ . K3 formality and the circle-extension model give the rational CDGA

$$\mathcal{A}_{q79} = H^*(B; \mathbb{Q}) \otimes \Lambda(u, v), \quad |u| = |v| = 1, \quad du = \delta, \quad dv = 0 \quad [9, 11].$$

The source-specific calculation uses the full intersection lattice $3U \oplus 2E_8(-1)$, not a fitted list of Betti numbers [30]. There are $24 \cdot 2 \cdot 2 = 96$ rational basis states. This is not the physical 96-dimensional Dirac carrier merely because the numbers agree. The generator v records the shared circle topologically, not its connection, charge spectrum or Lorentzian time.

Write t for the K3 top class, with $xy = (x \cdot y)t$ in degree four, and put

$$\gamma = \frac{\delta}{\delta^2} = -\frac{1}{4}\delta, \quad x_\perp = x - \frac{\delta \cdot x}{\delta^2}\delta.$$

On the factor without v , the source projection p and homotopy h are

z	1	x	t	u	ux	ut
$p(z)$	1	$x_\perp$	0	0	$ux_\perp$	ut
$h(z)$	0	$\frac{\delta \cdot x}{\delta^2}u$	$u\gamma$	0	0	0

and extend by tensoring with the identity on $\Lambda(v)$. Here p is the ambient idempotent onto representatives; factoring it through its image gives the usual projection and inclusion. They obey

$$dh + hd = I - p, \quad p^2 = p, \quad h^2 = ph = hp = 0.$$

This orthogonality is for the indefinite K3 intersection form, not a selected positive HYM inner product.

Since $\dim \delta^\perp = 21$, the retained representatives are $1, v, x, ux, xv, uxv, ut, utv$, with $x \in \delta^\perp$. Thus

$$(b_0, \dots, b_6) = (1, 1, 21, 42, 21, 1, 1), \quad \dim H^*(X_{q79}; \mathbb{Q}) = 88.$$

The smaller algebra generated by H, δ, u, v has 16 source states and only eight cohomology states, of ranks $(1, 1, 1, 2, 1, 1, 1)$. It is a submodel, not the full compactification. The cyclic pairing is the coefficient of tuv in a product. It is nondegenerate but not a positive Hilbert norm.

10.4 Strict topology does not mean a free physical theory

For these representatives, $m_2(a, b) = p(ab)$. A nonzero $h(ab)$ is proportional to $u\gamma$, possibly with v . Its product with a retained $\delta^\perp$ class vanishes because $\gamma \cdot \delta^\perp = 0$; its other products project to zero. Both terms of the ternary transfer therefore vanish:

$$m_3 = 0.$$

All homotopy images contain u , their pairwise products vanish, and another homotopy annihilates a product already containing u . These facts also kill every binary transfer tree of arity at least four. The source record confirms all $88^3 = 681472$ triples and gives the structural all-arity argument. Hence, at this topology-only tier,

$$m_1 = 0, \quad m_2 \text{ survives}, \quad m_n = 0 \quad (n \geq 3).$$

Strict transferred operations do not make the displayed representative inclusion multiplicative. For example,

$$p(H) = H, \quad HH = 2t, \quad p(HH) = 0, \quad h(HH) = 2u\gamma.$$

The homotopy carries the difference. Nor does algebraic p supply a spectral gap, an analytic Hodge projector or a low-energy truncation.

This result identifies where additional structure is needed, not a no-go for physical interactions. It concerns higher operations transferred from this particular rational cup-product DGA. A nonlinear residual with nonzero higher jets, twisted bundle coefficients, Yoneda operations or analytic overlap data is a different source of vertices. The repair-jet mechanism already explained in [section 11](#) remains valid; its physical coefficients must still be evaluated. The full Hull–Strominger source is not proved formal, and no spacetime scattering amplitude vanishes merely because the topology model has $m_3 = 0$.

The contrast with the auxiliary Heisenberg model is deliberate: $dc = ab$ leaves a nonzero ternary operation there; the selected q79 Hirsch contraction does not. The nonminimal CBF response hierarchy is a third object. None should replace either of the others solely because it is finite-dimensional.

10.5 How the compiler uses these results

The hidden projective rank-nine module contributes an honestly descending traceless adjoint of rank $9^2 - 1 = 80$. This fits the declared $3 + 8 + 8 + 80 + 3 = 102$ deformation block, but does not evaluate its endpoint arrays. The twisted coefficient model on the relative Jacobian J and the Hirsch model on X_{q79} cannot simply be tensored as if they already lived on the same differential-geometric source. A typed realization between those presentations and the metric/domain comparison of the preceding projection section are required.

For a supplied isometry U , operations transport exactly to its intrinsic range:

$$\widetilde{M}_n(y_1, \dots, y_n) = UM_n(U^\dagger y_1, \dots, U^\dagger y_n).$$

At arity zero this means $\widetilde{M}_0 = UM_0$. The unit, cyclic functional and pairings must also be transported to that range. Using an unrelated ambient unit or ambient rank-normalized trace is not the same construction. This is consistent with the corrected embedded-intertwiner and whole-projector results above.

Thus two obligations are resolved at their stated tiers: projective coefficient descent and rational q79 topology transfer. A selected physical coefficient-to-carrier map, common connection and analytic domains, signed action, and independent map to the accepted 27-state algebra remain distinct tasks. No numerical dimension equality closes those maps.

11 Higher repair jets and graph shadows

The operator is only the quadratic shadow of nonlinear repair. Higher derivatives of the same cost provide its formal interaction tensors.

Let E and F be finite-dimensional inner-product spaces, let $\Phi : E \rightarrow F$, and allow a positive residual metric field $W(a)$. At a background a_* , write

$$R_m = D^m \Phi(a_*), \quad W_m = D^m W(a_*), \quad V_n = D^n \mathcal{C}(a_*),$$

where

$$\mathcal{C}(a) = \frac{1}{2} \langle \Phi(a), W(a) \Phi(a) \rangle.$$

Theorem 11.1 (Complete repair-jet formula). *For every ordered three-color partition $[n] = A \sqcup B \sqcup C$, preserving the original argument order within each block,*

$$V_n(v_1, \dots, v_n) = \frac{1}{2} \sum_{A \sqcup B \sqcup C = [n]} \langle R_{|A|}(v_A), W_{|B|}(v_B) R_{|C|}(v_C) \rangle. \quad (11.1)$$

At a zero-defect background,

$$V_2(v, w) = \langle R_1(v), W_0 R_1(w) \rangle.$$

If W is constant and Φ is polynomial of degree r , then $V_n = 0$ for $n > 2r$.

Proof. Apply the multivariable Leibniz rule to the three factors Φ, W, Φ . Each derivative label is assigned exactly once to the left residual, metric, or right residual. Summing over all assignments gives (11.1). If $R_0 = 0$, the terms with an empty left or right residual block vanish at order two. Polynomial termination is immediate from the maximum derivative order of the two residual factors. $\square$

For the Maurer–Cartan residual $F(a) = da + a^2$ with constant metric, the repair cost therefore has only quadratic, cubic, and quartic vertices. They are not independent knobs: they are derivatives of one residual and one metric.

Once the reduced inverse G of V_2 is declared, Wick contraction organizes the formal coefficients by graphs [50]. The modal diagrammatics paper proves this graph expansion and its compatible projection theorem [44]. The present theorem adds the missing upstream arrow:

$$(\Phi, W, a_*) \longmapsto (V_2, G, \{V_n\}_{n \geq 3}) \longmapsto \text{formal repair graphs}.$$

This explains why Gaussian and graph structures recur. Every sufficiently smooth local repair process has a quadratic tangent shadow, and every formal quadratic expansion admits Wick organization. What is universal is the grammar. A physical Feynman rule additionally needs a signed action, grading, gauge complex, state, causal prescription, and renormalization.

12 Why repair is not automatically action

This section is not a caveat added after the fact. It is a theorem-level separation. Without it, the preceding construction would overclaim exactly where it becomes most physically interesting.

12.1 The variational-anchor test

Let E, F be finite-dimensional real Hilbert spaces and $\Phi : E \rightarrow F$. A constant variational anchor is a map $A : F \rightarrow E^*$ defining the one-form

$$\alpha_a(v) = (A\Phi(a))(v).$$

Theorem 12.1 (Finite Helmholtz gate). *On a star-shaped domain, a scalar functional S_{var} with $dS_{\text{var}} = \alpha$ exists if and only if α is closed, equivalently*

$$D(A\Phi)(a) = D(A\Phi)(a)^T \quad (12.1)$$

for every a . When it exists, one reconstruction is

$$S_{\text{var}}(a) = S_{\text{var}}(0) + \int_0^1 \alpha_{ta}(a) dt. \quad (12.2)$$

Proof. Exact one-forms are closed. Conversely, (12.2) is the Poincare radial homotopy on a star-shaped domain; differentiation and $d\alpha = 0$ give $dS_{\text{var}} = \alpha$. The coordinate form of closedness is (12.1). $\square$

In local field theory the same question becomes the graded formal-adjoint Helmholtz problem on jet bundles, with boundary terms and operator domains retained explicitly [47]. A positive metric on residual space does not supply this anchor.

12.2 The universal first-order lift

Even when a direct anchor is absent, introduce a multiplier $\lambda \in F$ and define

$$S_{\text{mult}}(a, \lambda) = \langle \lambda, \Phi(a) \rangle. \quad (12.3)$$

Theorem 12.2 (Multiplier lift and normal square). *Every exact closure state a_* gives a lifted critical point $(a_*, 0)$. If $J = D\Phi(a_*)$, the Hessian is*

$$\mathcal{D}_J = \begin{bmatrix} 0 & J^\dagger \\ J & 0 \end{bmatrix}, \quad \mathcal{D}_J^2 = \begin{bmatrix} J^\dagger J & 0 \\ 0 & J J^\dagger \end{bmatrix}. \quad (12.4)$$

Moreover,

$$\ker \mathcal{D}_J = \ker J \oplus \ker J^\dagger.$$

Proof. Varying (12.3) gives $\Phi(a) = 0$ and $D\Phi(a)^\dagger \lambda = 0$. Differentiating again at $(a_*, 0)$ gives the off-diagonal block operator. Direct multiplication gives its square and kernel. $\square$

The multiplier lift is a genuine first-order square root of the repair normal operator. It does not decide whether the multiplier is an ordinary field, response field, antifield, or unphysical auxiliary variable. That typing belongs to a later physical construction, such as a selected BV complex [45].

12.3 Information erased by the square

Proposition 12.3 (Normal-square nonselection). *The positive operator $J^\dagger J$ does not determine the sign or phase of J , the signed action Hessian, a causal inverse, or the first-order response. In particular,*

$$J_+ = \text{diag}(2, 3), \quad J_- = \text{diag}(-2, 3)$$

have the same normal square $\text{diag}(4, 9)$, but their inverse responses differ in the sign of the first channel.

If a signed self-adjoint action Hessian H_{var} is identified with J , then the repair Hessian is H_{var}^2 in the simplest equal-metric case. Squaring erases the Morse index. A saddle direction of the physical action becomes a positive repair direction.

The interaction vertices also differ. If

$$S_{\text{var}}(x) = \frac{m}{2}x^2 + \frac{g}{6}x^3, \quad \Phi(x) = S'_{\text{var}}(x) = mx + \frac{g}{2}x^2,$$

then

$$S_{\text{rep}}(x) = \frac{1}{2}\Phi(x)^2 = \frac{m^2}{2}x^2 + \frac{mg}{2}x^3 + \frac{g^2}{8}x^4.$$

The signed-action rows $(V_2, V_3, V_4) = (m, g, 0)$ become the repair rows $(m^2, 3mg, 3g^2)$. Repair graphs are therefore not physical Feynman graphs unless a separate theorem identifies the physical perturbative datum with the repair functional.

12.4 When a signed action does exist

Let $\mathcal{A}$ be a differential graded associative algebra with a degree-three cyclic trace τ satisfying

$$\tau(dx) = 0, \quad \tau(xy) = (-1)^{|x||y|}\tau(yx).$$

For $a \in \mathcal{A}^1$, define

$$S_{\text{MC}}(a) = \frac{1}{2}\tau(a da) + \frac{1}{3}\tau(a^3). \quad (12.5)$$

Theorem 12.4 (Cyclic Maurer–Cartan action). *For every degree-one variation u ,*

$$DS_{\text{MC}}(a)[u] = \tau(u(da + a^2)). \quad (12.6)$$

If the pairing between degrees one and two is nondegenerate, the critical locus is exactly the Maurer–Cartan locus. The action is infinitesimally gauge invariant under $\delta_\epsilon a = d\epsilon + [a, \epsilon]$. It descends through even projective overlaps because the scalar cocycle cancels under conjugation and the cyclic trace is conjugation invariant.

Proof. Differentiate the quadratic term. The Stokes identity moves d from the variation to a , making the two contributions equal. Cyclicity makes the three cubic contributions equal. This gives (12.6). Pairing a gauge variation with the residual and integrating by parts yields the Bianchi identity. On overlaps, each monomial is conjugated by the same even transition, and the trace removes that conjugation. $\square$

On a compact complex threefold with holomorphic volume form Ω , the candidate trace is

$$\tau(x) = \int_X \Omega \wedge \text{Str}(x),$$

and (12.5) becomes holomorphic Chern–Simons theory [52, 8]. This supplies a signed action for the integrability lane. It does not automatically supply nondegenerate cyclic pairings for the HYM, balanced, anomaly, coframe, and Lorentzian lanes.

13 The cohesive closure-repair theorem

The preceding results can now be stated as one composition theorem.

Theorem 13.1 (Cohesive closure-repair and shadow theorem). *Suppose the following data are supplied:*

1. *a metric configuration space with a smooth defect section Φ , smooth metric data and an exact-zero background p_* (a C^N source gives only the cost jet through the available order);*
2. *compatible local isometric/unitary descent data;*
3. *on the cohesive branch, an integrable possibly twisted superconnection whose endomorphism algebra realizes the defect complex;*
4. *a Hermitian structure and self-adjoint elliptic domains; and*
5. *any declared orthogonal projection or isometric transform reducing the linearized operator.*

Then one and the same source determines, functorially:

$$\begin{aligned} \mathcal{C} &= \tfrac{1}{2} \|\Phi\|^2, & X &= -\text{grad } \mathcal{C}, & H &= D\Phi(p_*)^\dagger D\Phi(p_*), \\ e^{-sH}, & P_0, & G &= H^{-1}(I - P_0), & \chi_I(H), & K_f(x, y) = \langle x | f(H) | y \rangle, \\ & \{D^n \mathcal{C}(p_*)\}_{n \geq 3} & & & & \text{and their finite formal repair graphs.} \end{aligned} \quad (13.1)$$

The nonlinear flow, tangent calculus, and existing basin limits descend through the compatible local chart equivalences. Reducing projections transport the tangent functional calculus. An onto unitary equivalence of the full residual and metric transports every object in (13.1). An embedded nonlinear subsystem additionally needs invariance of its repair vector field; exact retained graph transport requires compatibility of the full typed vertex, grading and contraction data, as in the diagrammatics theorem. Reduction of the quadratic operator alone does not supply these nonlinear hypotheses.

The theorem does not identify $\mathcal{C}$ with a signed physical action. Such an identification requires a variational anchor or a separately typed multiplier/cyclic construction. Nor does the theorem select the original defect, metric, background, causal domain, state, or projection.

Proof. The first variation, full Hessian, and exact-zero normal square are theorems 2.1 and 2.2. The repair flow and its tangent semigroup are theorem 3.1. Cohesive realization and twist cancellation are theorems 4.1 and 4.2. Elliptic Hodge theory supplies the functional calculus in theorem 6.1. Nonlinear descent and compatible transport are theorems 7.1 and 7.2. The complete cost jet is theorem 11.1; formal Wick contraction then produces its graph weights. The final nonidentification statement follows from theorems 12.1 to 12.4. $\square$

The theorem says something stronger than “an operator generates repair.” The nonlinear repair field exists first. Its derivative produces the operator. Yet it also says something more careful than “closure explains all operators.” Only the displayed class of positive tangent operators is obtained, and only after its source data are supplied.

14 The q79 benchmark and its exact boundary

14.1 What is already constructed

The current q79 research program provides a hash-bound α -twisted cohesive benchmark $\mathcal{S}_{\text{HS}}$ on J . On that object, the following are exact:

1. the twisted cohesive representative and ordinary global endomorphism dg algebra;
2. the canonical Maurer–Cartan curvature residual;
3. the positive Hodge repair Hessian after a Hermitian choice;
4. the tangent heat semigroup, harmonic projector, and reduced Green operator on the global total space;
5. projective local descent of the finite coherent repair witness; and
6. conditional transport of Ext, Yoneda, and formal Maurer–Cartan data through a dg-enhanced BHT/Fourier–Mukai equivalence.

The q79 result is not merely a finite analogy. It applies the general mathematics to one explicit twisted perfect-complex atlas. But it is a benchmark object, not yet the selected physical visible/hidden endpoint.

14.2 The proposed full defect and why it remains proposed

The physical source program organizes the Hull–Strominger obligations into a multi-lane residual of the schematic form

$$\Phi_{\text{phys}} = (\Phi_0, \mu_{TX}, \mu_V, \mu_W, \Phi_{\text{bal}}, \Phi_{\text{AB}}, \Phi_{\text{SU}(3)}). \quad (14.1)$$

Here Φ_0 denotes integrability, the μ -rows denote HYM moment maps, Φ_{bal} balanced geometry, Φ_{AB} anomaly/Bianchi closure, and the final row the declared $\text{SU}(3)$ normalization or coordinate constraint.

If a selected endpoint s_* satisfies $\Phi_{\text{phys}}(s_*) = 0$, then the positive physical repair Hessian is exactly

$$A_* = D\Phi_{\text{phys}}(s_*)^\dagger D\Phi_{\text{phys}}(s_*). \quad (14.2)$$

Writing the linearization as a deformation-complex block J_* and additional physical rows K_* gives

$$A_* = J_*^\dagger J_* + K_*^\dagger K_*. \quad (14.3)$$

This is an exact identity *if the same selected endpoint, metric, and domains supply every row*. The current corpus has not yet supplied that complete physical object. Equation (14.3) is therefore the correct target theorem, not a declaration that the target has already been reached.

14.3 The explicit residual and its block support

The full-residual record improves the schematic notation in (14.1) to an explicit differential-operator compiler. Write $\nu = \|\Omega\|_\omega$, and reserve c_{vol} for the volume normalization constant, not a third Chern class. The six extra rows are

$$\begin{aligned}\mu_{TX} &= R_\Theta \wedge \omega^2, & \mu_V &= F_V \wedge \omega^2, & \mu_W &= F_W \wedge \omega^2, \\ \mathcal{B} &= d(\nu\omega^2), \\ \mathcal{A} &= dH - \frac{\alpha'}{4}(\text{tr } R_\Theta^2 - \text{tr } F_V^2 - \text{tr } F_W^2), \\ \mathcal{N} &= i\Omega \wedge \bar{\Omega} - c_{\text{vol}}\omega^3.\end{aligned}\tag{14.4}$$

Their target is the orthogonal sum of the six spaces

$$\begin{aligned}\Omega^6(\text{ad } TX), & \quad \Omega^6(\text{ad } V_3), & \quad \Omega^6(\text{ad } W_9^\tau), \\ & \quad \Omega^5, & \quad \Omega^4, & \quad \Omega^6.\end{aligned}$$

This is a choice of positive repair metric after a common scale is factored out. It is not an identification with a Lorentzian action.

At a fixed complex-structure chart, put $a = (a_{TX}, a_V, a_W)$. Exterior differentiation and the covariant curvature variation give

$$\begin{aligned}K_{TX}v &= d_\Theta a_{TX} \wedge \omega^2 + 2R_\Theta \wedge \omega \wedge \dot{\omega}, \\ K_Vv &= d_{A_V} a_V \wedge \omega^2 + 2F_V \wedge \omega \wedge \dot{\omega}, \\ K_Wv &= d_{A_W} a_W \wedge \omega^2 + 2F_W \wedge \omega \wedge \dot{\omega}, \\ K_Bv &= d\{\nu(2\omega \wedge \dot{\omega} + \log \dot{\nu} \omega^2)\}, \\ K_{\mathcal{A}}v &= d\dot{H} - \frac{\alpha'}{2}\{\text{tr}(R_\Theta \wedge d_\Theta a_{TX}) - \text{tr}(F_V \wedge d_{A_V} a_V) - \text{tr}(F_W \wedge d_{A_W} a_W)\}, \\ K_{\mathcal{N}}v &= i(\dot{\Omega} \wedge \bar{\Omega} + \Omega \wedge \dot{\bar{\Omega}}) - 3c_{\text{vol}}\omega^2 \wedge \dot{\omega}.\end{aligned}\tag{14.5}$$

Here $\log \dot{\nu} = \text{Re}\langle \dot{\Omega}, \Omega \rangle / \|\Omega\|^2 - \frac{1}{2} \text{tr}_g \dot{g}$ in the packet's fixed-chart metric convention. Varying complex structure requires the corresponding typed chart terms as well; these formulas do not supply their endpoint coefficients. In particular the factor $\alpha'/2$, rather than $\alpha'/4$, is forced by differentiating the curvature square.

The rank-102 complex fiber is

$$\mathcal{Q} = T^*X \oplus \text{ad}(TX) \oplus \text{ad}(V_3) \oplus \text{ad}(W_9^\tau) \oplus TX, \quad \mathbf{r} = (3, 8, 8, 80, 3).$$

It is not the full form-augmented Hilbert space and not a finite discretization dimension. In this lane order the extra-row incidence is

$$I_K = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

A one means allowed dependence, not a certified nonzero coefficient. The anomaly row has a common target for all five lanes. Its Gram square therefore allows all 25 ordered blocks

and $102^2 = 10\,404$ ordered fiber positions. Omitting it had allowed only 19 blocks and 7716 positions. The six cross-gauge blocks contribute $2(8 \cdot 8 + 8 \cdot 80 + 8 \cdot 80) = 2\,688$ positions. They must not be forced to zero before evaluating the same-source operator. Nor may the real Bianchi row be deleted as already contained in the complex without a same-domain isometry proving that exact redundancy.

This correction changes the possible coupling pattern, not the positivity identity. On the common closed form domain,

$$H_{\text{phys}} = \Delta_{\mathcal{Y},1} + K^\dagger K \geq \Delta_{\mathcal{Y},1}, \quad \ker H_{\text{phys}} = \ker \Delta_{\mathcal{Y},1} \cap \ker K.$$

The finite witness in the source has $\Delta = \text{diag}(0, 0, 1, 2, 4)$ and $K = (1, 1, 0, 0, 0)$. It removes one of two harmonic directions and retains a positive gap of one. That example shows why neither the harmonic projector nor a scalar rescaling of the bare Hodge operator can be assumed.

There is also a lawful conditional sparsification. If the physical endpoint, residual, metric and Galerkin projection all respect the projective $E[3]$ action, character orthogonality removes unequal-character blocks. The hidden adjoint multiplicities are 8 for the trivial character and 9 for each of eight others. If, additionally, all 22 non-hidden modes are trivial, the count becomes

$$30^2 + 8 \cdot 9^2 = 1\,548.$$

The general count is $\sum_{\chi} (m_{\text{ext},\chi} + m_{\text{hid},\chi})^2$. The value 1 548 is not a selected physical matrix count until these representation and invariance hypotheses are verified. It complements, and does not retract, the unconditioned 25-block support.

These are imported compiler results, not newly proved source existence. The earlier candidate audit in that record is historical. In particular, later hidden-carrier and existential HYM progress retained in the Flux paper is not undone by its old status fields. A common physical visible endpoint, all differential residual rows and their numerical coefficients are the stronger input needed here.

14.4 The shared circle

The shared circle enters this construction as a differential line $(L_{\text{sh}}, \nabla_{\text{sh}})$ and its equivariance, not as an extra coordinate and not as physical time. The finite q79 theorem already shows that one universal flat cyclic line acts compatibly on the selected finite carriers and commutes with their projectors and finite Hessian. The continuum promotion must prove the corresponding relation

$$[\nabla_{\text{sh}}, D\Phi_{\text{phys}}] = 0$$

in the appropriate equivariant sense. It would then follow that the repair Hessian and all its functional-calculus shadows preserve the shared charge sectors. This is a serious unification condition. It does not identify compact phase with Lorentzian time.

14.5 Status table

Object or arrow	Status	Boundary
General covariant residual Hessian	Exact here	Includes nonzero-defect correction
Exact-zero normal square and repair semigroup	Exact here	Requires selected metric and domain
Twisted cohesive endomorphism dg algebra on $\mathcal{S}_{\text{HS}}$	Exact structural benchmark	Scalar twist cancels in End
Maurer–Cartan repair and Hodge tangent on $\mathcal{S}_{\text{HS}}$	Exact structural benchmark	Hermitian metric exists but is not physically selected
Local nonlinear repair descent	Exact here	Does not select one basin
Isometric functional-calculus transport	Exact conditional	Arbitrary derived equivalence is not unitary
Repair jet to formal graph data	Exact finite/formal	Not a physical QFT action
Cyclic action for the integrability lane	Exact conditional	Requires cyclic trace and domain hypotheses
Selected physical V_3/W_9 cohesive or pure-bundle endpoint	Open	Current source-selection frontier
Full HYM/balanced/anomaly action and real slice	Open	Current action frontier
Lorentzian causal operator and state	Open	Cannot be inferred from the positive square
Continuum-to-finite product/Hodge intertwiner	Open	Current geometry frontier

15 What the theorem changes downstream

15.1 Fixed points

The fixed-point theory keeps its existing theorem ownership. It proves existence, uniqueness under suitable hypotheses, smoothing, projection control, and convergence. The present paper supplies one principled source for the energy and tangent generator to which those theorems may be applied. It does not repeat their analytic hypotheses or promote a finite witness to an infinite-dimensional flow.

15.2 Wave and particle representations

The positive filtered operator in the wave–particle paper no longer needs to be viewed as mathematically arbitrary on the cohesive benchmark. It can be the functional calculus of a repair Hessian. Its local and spectral descriptions then become the localized and normal-mode shadows of one tangent repair process. The quantum state, Born bridge, instrument, and physical time evolution remain separate obligations.

15.3 Perturbative graphs

The graph paper begins with typed quadratic and higher tensors. The repair jet theorem derives one entire such tensor family from (Φ, W) . This removes independent repair vertices as source inputs. It does not remove physical couplings from a QFT until the repair functional is proved to be the physical signed action or is related to it by a certified variational map.

15.4 Quantum field theory

The existing selected free q79 CAR/AQFT result uses standard quantum-field machinery after its geometric input is supplied. The present paper does not rederive that machinery. It clarifies the upstream division of labor: positive repair controls stabilization and Euclidean/Hodge localization; the signed gauge-fixed action controls physical response and causal Green operators; the state and renormalization prescription complete the quantum theory.

15.5 Gravity

If a gravitational defect system is selected, its positive repair Hessian and kernel follow by the same theorem. This supports the MTT intuition that geometry may respond to closure pressure. It does not prove that Einstein's equations are the selected repair equation, that the stress tensor has the required normalization, or that the resulting theory is UV complete. The Lorentzian principal symbol and causal domains must be derived from the signed physical system, not guessed from a positive Hilbert-space metric.

16 Theorem ownership and corpus placement

This paper is intended to prevent the same upstream theorem from being reproved in several downstream papers.

Result	Canonical owner	Use elsewhere
Covariant residual Hessian and exact-zero normal square	This paper	Foundations, Fixed Points, QG may cite
Repair flow to tangent heat semigroup	This paper	Wave-particle and stabilization papers may cite
Cohesive q79 realization and twist cancellation	This paper as synthesis; executable theorem packet as evidence	Source and bundle papers may cite
Repair descent and compatible functional calculus	This paper	Projection papers may import conclusion
Repair jet to formal graph tensors	This paper	Modal diagrammatics retains graph theorem ownership
Finite graded Wick and graph-projection theorem	Modal diagrammatics	Imported here without proof duplication
Fixed-point existence and convergence	Fixed Points I and successors	Imported here
Conditional wave-particle operator encoding	Wave-particle paper	Receives a structural source candidate here
Physical q79 endpoint and action	No completed owner yet	Research frontier, not a theorem in this paper

The new result should therefore be added to the related papers by a concise cross-reference and status update, not by copying the proofs. This keeps the corpus readable and makes later corrections local.

17 Falsifiers and acceptance tests

The structural theorem is falsified in an application if any required map or domain fails. The physical interpretation fails if a stronger claim is made without its bridge. A candidate

promotion should pass the following tests.

1. **Defect source:** specify every residual row, bundle, connection, domain, and source hash.
2. **Background:** prove whether the selected state has zero defect. If it does not, compute the residual Hessian correction.
3. **Metric:** derive or explicitly declare the field and residual metrics; do not infer them from a categorical equivalence.
4. **Descent:** verify the literal overlap maps, cocycles, and isometric/unitary compatibility.
5. **Hodge domain:** prove self-adjointness, ellipticity or the appropriate hyperbolic replacement, and boundary conditions.
6. **Projection:** verify reduction of the operator and preservation of the nonlinear residual jet, not only matching dimensions.
7. **Action:** provide a Helmholtz/cyclic anchor or a fully typed multiplier/BV lift. Do not call the positive square the physical action.
8. **Causality:** derive retarded, advanced, or Feynman prescriptions from the signed gauge-fixed operator and state.
9. **Strata:** test Green and contraction bounds near every cohomology-rank jump.
10. **Numerics:** evaluate endpoint coefficients only after the same source has passed the structural tests.

These tests turn the paper into a research instrument. A failure identifies which arrow is missing; it does not send the program back to the beginning.

18 Frontier theorem

The later endpoint-factorization record organizes seven acceptance rows into three structured sources:

1. **GAS**, geometry and action: the physical endpoint and signed upper action;
2. **SYN**, spectral synthesis: the analytic intertwiner and compatible chain/product contraction data;
3. **BV4**, a BV-compatible four-dimensional compactification.

Shared-line functional calculus and the Galerkin/Feshbach execution are then consequences of compatible GAS and SYN data, rather than two more independent source objects. Three structured sources are not three scalar parameters. None of the three complete physical packages is accepted in the cited record. This dependency reduction preserves the exact finite constructions while preventing another round of inventing independent kernels for already derived arrows.

The decisive next theorem is not another abstract Hessian statement. That part is now complete. It is a selected-source theorem of the following form.

Conjecture 18.1 (Selected physical cohesive-source theorem). There exists a selected q79 Hull–Strominger endpoint s_* , represented either by a stable pure-bundle source or by a physically admitted cohesive source, together with:

1. the full defect section Φ_{phys} and a common Hermitian metric;
2. the shared differential line and equivariant connections;

3. a cyclic or BV variational structure whose Euler–Lagrange rows agree with the selected physical residual on a declared gauge slice;
4. Lorentzian externalization, real structure, and causal domains; and
5. a continuum-to-finite chain, product, Hodge, and normalization intertwiner with an error certificate.

At an exact endpoint, the positive repair operator is $A_* = D\Phi_{\text{phys}}^\dagger D\Phi_{\text{phys}}$, while the signed physical operator is obtained from the certified variational structure rather than inferred from A_* .

Proving this conjecture would connect the upper bundle geometry, cohesive deformation theory, shared phase, fixed points, wave kernels, finite q79 operators, and the signed physical theory without conflating them. It is a sharp target because every required object has a declared type and every invalid shortcut has already been removed.

19 Discussion

19.1 What has genuinely been gained

The main gain is upstream organization. The heat kernel is no longer merely an attractive analogy for coherence. Under the theorem’s hypotheses it is the exact tangent propagator of nonlinear defect repair. The harmonic sector is the kernel of the same Hessian. The spectral projector and local kernel come from its functional calculus. The formal repair vertices are higher derivatives of the same cost. Compatible projection preserves the whole package.

This makes several previously separate papers part of one coherent story:

$$\begin{aligned}
 &\text{closure defect and metric} \longrightarrow \text{repair flow,} \\
 &\text{repair flow} \longrightarrow \text{Hodge Hessian,} \\
 &\text{Hodge Hessian} \longrightarrow \text{heat, Green, and spectral kernels,} \\
 &\text{higher repair jet} \longrightarrow \text{formal graph grammar,} \\
 &\text{compatible projection} \longrightarrow \text{retained shadows.}
 \end{aligned}$$

19.2 What has not been gained

The same clarity exposes the missing physics. A positive normal square cannot tell us which first-order sign or phase nature uses. It cannot choose a Lorentzian causal prescription. It cannot distinguish a physical saddle from a repair minimum. It cannot supply a quantum state, a Born rule, or a renormalization scheme. These are not small numerical corrections; they are different typed structures.

The result is therefore neither “only mathematics” nor a completed theory of everything. It is a rigorous mathematical bridge that narrows the physical frontier. It tells us which downstream constructions are already forced once a selected source exists, and which ones still require genuinely new input.

19.3 Why the cohesive route matters

The cohesive route is especially useful because it remains regular where an open-stratum bundle contraction fails. It allows the full derived object to carry a global differential, deformation algebra, and Hodge calculus without first pretending that the cohomology rank

is constant. This may be the right preprojection language for MTT. It may also turn out to be only an auxiliary resolution of a physical pure-bundle endpoint. The present theorem permits both outcomes and gives a concrete test between them.

20 Conclusion

Closure is not represented here first as a static operator. It is represented by a nonlinear defect and the process that reduces its cost. The operator appears as the derivative of that process at an admissible state. Hodge theory, heat flow, Green response, spectral projection, local kernels, and formal repair graphs then follow as exact shadows of one source.

The result is strongest precisely because its boundary is explicit. At a nonzero-defect critical point the Hessian contains a residual correction. A derived equivalence is not automatically isometric. A global cohesive operator does not guarantee a uniform fiberwise Green family. A positive repair functional is not a signed action. A kernel is not automatically a quantum propagator. A descended basin map is not automatically unique actualization.

For q79, the cohesive Hartshorne–Serre benchmark now realizes the structural chain exactly after a Hermitian choice. The next leap is no longer to guess another operator. It is to select the physical endpoint, metric, cyclic or BV action, Lorentzian domains, and continuum-to-finite intertwiner from one source. If that is achieved, much of the downstream operator language will already be waiting as theorem rather than assumption.

Computational Evidence and Reproducibility

The universal proofs in this paper are analytic or finite algebraic. Their MTT-specific instantiations are pinned in the curated [MTT Results Reproduction repository](#). The principal result records are:

- the q79 cohesive-superconnection record;
- the q79 Maurer–Cartan repair record;
- the local Cech–Hilbert and nonlinear repair-descent records;
- the covariant residual-Hessian record and nonzero-defect witness;
- the reversible–dissipative repair record;
- the repair-jet graph-functor record;
- the variational-anchor and multiplier-lift record;
- the cyclic Maurer–Cartan action-descent record; and
- the q79 string cyclic-cotangent boundary record.

Version 3 additionally imports:

- the explicit six-row/25-block compiler: `q79_v3w9_full_residual`;
- its conditional character reduction: `q79_rank102_qutrit_reduction`;
- the bounded transported-metric record: `preprojection_bht_transported_metric`;
- the endpoint factorization record, listed under `causal_base_q79_seven_row_endpoint_factorization_packet`.

Version 4 reviews `q79_continuum_finite_intertwiner_cutset`. Its exact intertwining conditions and rational witnesses are retained; its unqualified whole-projector gap estimate is corrected explicitly in the projection section. The frozen artifact remains unchanged for

audit. The local Cech equalizer, nonlinear descent, nonzero-defect example and phase-loss and damping witnesses are explicitly compared in the current contextual integration ledger. These imports use the frozen result manifest at [f141a20ea23c](#); the integration companion gives each artifact hash and manuscript location. Each record declares its theorem tier, source artifacts, verifier, zero-fit status, and nonpromotion guards. The exact witnesses use rational or symbolic arithmetic where stated. They do not select the physical q79 endpoint or supply measured values. The source TeX, generated Markdown, compiled PDF, metadata hashes, and version-provenance audit accompany this paper in the public MTT papers repository. The evidence snapshot used for Version 1 is commit [ba0874012246bb930677ebfa6f0a6eb33a57a8f0](#).

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